Design Science

Tolerancing in product design – a holistic description scheme

Stefan Goetz [1], Martin Roth [1,2], Paul Schaechtl [1], Vincent Kramer [1], Christoph Bode [1], Dennis Horber [1], Michael Franz [1], Stephan Freitag [1], Jörg Miehling [1] and Sandro Wartzack [1]

[1]*Engineering Design, Friedrich-Alexander-Universität Erlangen-Nürnberg, Erlangen, Germany*
[2]*Department of Industrial and Materials Science, Chalmers University of Technology, Gothenburg, Sweden*

Abstract

Along with increasing product complexity and quality requirements, the consistent consideration of inevitable production-induced variations within the product development process becomes a decisive factor for the market success of products. Consequently, various tolerancing approaches have emerged over the last few decades. However, tolerancing is considered complex in education, research, and industry, as it is a highly interdisciplinary task that takes place at different levels of detail, ranging from Robust Design to tolerance specification to manufacturing process design. This contribution proposes a novel approach that allows a holistic and structured description of tolerancing and fosters a common understanding among all involved stakeholders from design, manufacturing, and inspection. This is achieved by categorizing it into several distinct elements: activities, methods, tools, models and data, information, and knowledge. This ensures clarity and supports the utilization of existing approaches. While this contribution focuses on tolerancing in product design, the linkage to subsequent product realization stages and further engineering domains is also addressed in the proposed description scheme. An exemplary classification of research papers and a description of a practical development process of a technical system with its different tolerancing activities illustrate the benefits of the proposed description scheme.

Keywords: Tolerancing, Dimensional Management, Variation Management, Robust Design, Product Design

Received 23 August 2024
Revised 14 March 2025
Accepted 20 March 2025

Corresponding author
Stefan Goetz
goetz@mfk.fau.de

© The Author(s), 2025. Published by Cambridge University Press. This is an Open Access article, distributed under the terms of the Creative Commons Attribution licence (http://creativecommons.org/licenses/by/4.0), which permits unrestricted re-use, distribution and reproduction, provided the original article is properly cited.

Des. Sci., vol. 11, e11
journals.cambridge.org/dsj
DOI: 10.1017/dsj.2025.6

1. Introduction

Tolerancing has a long tradition, starting with the first dimensional tolerances in the early 1900s, followed by the standardization of Geometric Dimensioning and Tolerancing (GD&T), up to today's common computer-aided tolerancing, e.g., utilizing commercial simulation software and profiting from the recent advances along with Industry 4.0 (Hong & Chang 2002; Schleich *et al.* 2018). It describes the management of geometrical variations along the product life cycle, from the early design stages to process planning, manufacturing, and geometrical inspection (Dantan *et al.* 2008b) and is often used as a synonym for (geometrical) variation management (Dantan *et al.* 2008b; Schleich *et al.* 2018), dimensional management (Craig 1996; Nickolaisen 1999), or tolerance management (Graves & Bisgaard 1999; Anwer, Cid & Mathieu 2003). Thus, there are many different definitions and

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

distinctions (Schleich 2017). Accordingly, driven by the interdisciplinary nature of tolerancing, including engineering, manufacturing, and metrology, there are numerous research approaches, some of which have already been implemented in industry. Nevertheless, further research in tolerancing is required to overcome the challenges associated with new manufacturing technologies and virtual product development processes as well as to improve practical applicability (Morse *et al.* 2018). All of these existing and future tolerancing approaches share the same objective – they aim to efficiently assure the functionality and quality of the final product by considering and controlling the variations that inevitably occur during production and in use. Nevertheless, the vast array of tolerancing approaches addresses many diverse and, at times, intertwined aspects, which is partly shown in multiple review papers representing the state of the art from a specific point of view, e.g., for tolerance representation (Roy, Liu & Woo 1991; Qin *et al.* 2018), tolerance analysis (Nigam & Turner 1995; Singh, Jain & Jain 2009a; Marziale & Polini 2011; Schleich *et al.* 2014; Cao, Liu & Yang 2018; Dantan & Homri 2024), dimensional tolerancing (Voelcker 1998), standard-compliant tolerancing (Henzold 2021; Maltauro *et al.* 2024a), tolerance synthesis (Singh, Jain & Jain 2009b) automated tolerancing (Yu, Tan & Yuen 1994), optimal tolerancing (Creveling 1997; Armillotta 2020; Hallmann, Schleich & Wartzack 2020a), and tolerancing in product and process design (Ngoi & Ong 1998; Kumar, Soundararajan & Alagumurthi 2009; Karmakar & Maiti 2012), or giving an overview of tolerancing activities (Drake 1999; Morse *et al.* 2018; Wartzack 2024).

However, a global view on tolerancing providing a consistent, detailed description is missing. This goes along with a lack of differentiation and classification of the various existing approaches, which is impeded by their strong interactions, e.g., between tolerance analysis model setup and definition of the assembly process. Different analyses of existing tolerancing research (Garaizar *et al.* 2014; Krogstie *et al.* 2015) emphasize that this can lead to ambiguous interpretations of the current state of research and the derived future research directions. This becomes evident in the review of past tolerancing conferences, namely the CIRP Conference on Computer Aided Tolerancing (CAT), stating difficulties in the classification of approaches due to the diversity of keywords provided, thus hindering a systematic analysis of the associated papers (Garaizar *et al.* 2014) and finally leading to redundant work (Blessing & Chakrabarti 2009). Beyond research, this complicates the implementation process within the industry (Sigurdarson, Eifler & Ebro 2018).

Motivated to close this gap, this contribution proposes to understand tolerancing as a holistic system by providing a novel description scheme incorporating different viewpoints, objectives, and constraints by answering the following research question: *How can tolerancing be described more holistically and unambiguously?*

The paper is structured as follows: Section 2 discusses the related works on structuring and presenting tolerancing and defines the fundamental tolerancing activities. Section 3 presents the development of the novel scheme to describe tolerancing. Its initial evaluation is done by an exemplary categorization of research articles in Section 4.1 and its practical application to a use case in Section 4.2. After a final discussion on the benefits and limitations of the proposed approach in Section 5, the paper closes with a conclusion and outlook in Section 6.

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

2. Related works and fundamentals

To motivate the development of the intended description scheme and to differentiate it from related approaches, Section 2.1 discusses preliminary work on structuring tolerancing. Section 2.2 then presents a fundamental assignment of tolerance-related activities along the product life cycle, which forms the basis for the proposed scheme.

2.1. Structuring and presenting tolerancing

Different process models exist in industrial practice (Stockinger 2010; Schleich 2017) for organizing the diverse tolerancing activities. For example, in the automotive industry, tolerancing processes are described with seven to nine predefined steps (Bohn & Hetsch 2013; Mölzer & Strobelt 2013), which are transferred into a standardized process through the German Association of the Automotive Industry (VDA) (Verband der Automobilindustrie 2013). However, the focus is on a high organizational level and does not address specific aspects of implementing tolerance analysis, for example.

In contrast, several research projects have been carried out to develop a structured tolerancing process on a more specific level. Some focus on embedding tolerancing activities in early stages of the product development process (Roy *et al.* 1991; Islam 2009), e.g., by a structured translation of functional requirements into geometrical definitions in the concept design (Roy *et al.* 2001; Dantan, Anwer & Mathieu 2003), an approach enabling the earliest possible definition of geometry (Ballu *et al.* 2006; Costadoat *et al.* 2009; Costadoat, Mathieu & Falgarone 2011) or a combination of tolerancing and Robust Design activities (Hu & Peng 2007; Goetz, Schleich & Wartzack 2020). In addition to intensive frontloading, the aim is to achieve parallelization of design and tolerancing as well as consistency (Desrochers & Laperrière 2003; Dufaure & Tessandier 2008). Other approaches concentrate on the structure and detailed description of tolerancing activities as a process, highlighting the interdisciplinarity and variety of tolerancing approaches, e.g., by providing a SysML-based representation of a variation management process (Horber *et al.* 2022), a computer-aided toolchain enabling the geometry assurance from concept phase to verification and production (Söderberg, Lindkvist & Carlson 2006; Söderberg *et al.* 2016), a framework to consider perceived quality in tolerancing (Stylidis, Wickman & Söderberg 2020), a tolerance analysis and synthesis approach integrating non-geometrical functional relations (Germer & Franke 2003) and a description of computer-aided tolerancing from tolerance specification phase to product usage (Wartzack *et al.* 2011). The partly addressed iterative character is explicitly described in the closed-loop tolerance engineering model based on four basic recurring steps (Krogstie & Martinsen 2012; Schleich & Wartzack 2014). These iterative and cross-domain, cross-phase, and cross-activity processes lead to different levels of detail of tolerance information. To improve consistency and traceability, various unifying information models for the combination of these processes exist (Bradley & Maropoulosa 1998; Feng & Song 2000; Johannesson & Söderberg 2000; Sudarsan *et al.* 2000; Roy *et al.* 2001; Desrochers & Laperrière 2003; Ballu, Dufaure & Teissandier 2007; Mathieu & Ballu 2007; Shen, Shah & Davidson 2008; Dantan, Ballu & Mathieu 2008a; Zhang, Jiang & Ning 2009; Rhahli *et al.* 2010). Linking to this, the ISO GPS system can be used for consistent

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

communication of tolerances between the various actors along the product life cycle, explicitly addressing the link between specification and verification of tolerances (ISO/TC 213 2011, 2014).

While these approaches support engineers in the proper definition and communication of tolerance information or tolerancing procedures, they do not support the consistent description of tolerancing, especially when considering the various fields of application, stages and stakeholders along the product life cycle (Maltauro, Meneghello & Concheri 2024b), see also Figure 1. This also applies to the few approaches that assign tolerancing activities to individual product development process steps, for example, in the VDI 2221 (Leidich, Ebermann & Gröger 2014; Goetz 2022), with the aim of structuring tolerancing in product design. This is due to the fact that they focus only on certain aspects of tolerancing, such as early tolerance management (Goetz 2022). The heterogeneity of these individual approaches already emphasizes that there can be no generally valid and yet concrete tolerancing process, thus hindering the exchange in research and industry (Sigurdarson *et al.* 2018). This is further amplified by different views on tolerancing between distinct companies, departments, or researchers (Krogstie, Martinsen & Andersen 2014) and the interdisciplinary nature of tolerancing since similar methods, e.g., for tolerance analysis, are used in early Robust Design (Goetz, Schleich & Wartzack 2018a), final tolerance optimization (Hallmann, Schleich & Wartzack 2021b), and tolerancing-driven process design (Heling *et al.* 2019).

Similar problems are faced in the Robust Design domain. For this reason, a matrix-based classification framework differentiating between "System Design," "Parameter Design," and "Tolerance Design" on the one hand and between "Principles and Practices," "Methods and Tools," "Application Cases," and "Implementation" on the other hand is proposed (Eifler & Schleich 2021). Although this allows the mapping of Robust Design research and thus the identification of remaining research potentials, the classification process itself remains challenging since, for example, a clear distinction between contributions proposing a new method or its application is difficult (Eifler & Schleich 2021). This indicates that a simple classification with two dimensions inevitably leads to unambiguity, as most contributions address multiple aspects.

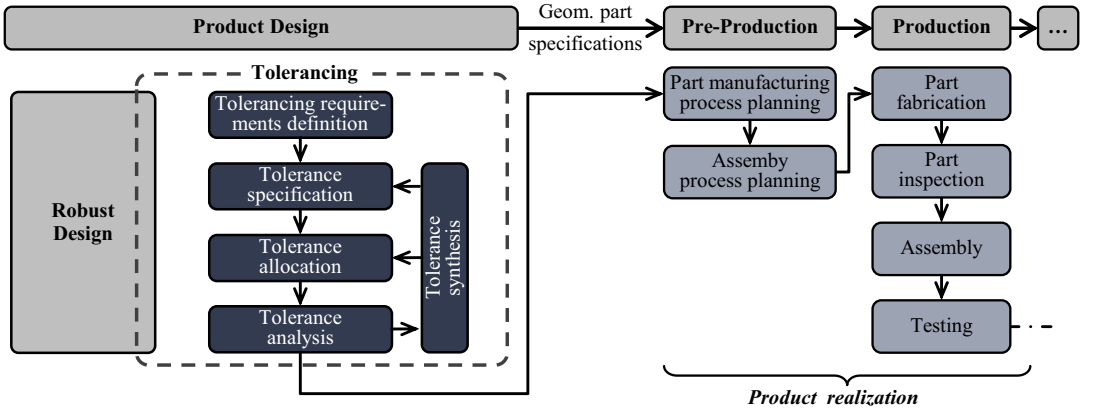


Figure 1. Tolerance-related activities along the product life cycle. Inspired by Taguchi et al. (2005), Schleich (2017) and Roth (2024)).

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

2.2. Fundamental tolerancing activities

Considering the various referenced works, Figure 1 provides an overview of the tolerance-related activities along several product design and realization stages. Product realization is to be understood as the product life cycle phases that convert the ideas from design into the real physical product, so it includes pre-production and production steps.

Following the definition in Schleich (2017) and focusing on the product design perspective, tolerancing can be traced back to the activities of tolerancing requirements definition, tolerance specification, tolerance allocation, tolerance analysis, and tolerance synthesis. These partly overlap with the activities of the late Robust Design, namely the tolerance design according to Taguchi *et al.* (2005). However, since most Robust Design activities, especially in the early system and parameter design, are assigned to the product design process itself and only have overlaps with tolerance activities, only the five key tolerancing activities, described in the following, are used as the basis for the proposed description scheme.

- **Tolerancing requirements definition:** The globally defined product requirements (from user demand, etc.) are broken down into a set of quantitative tolerancing requirements that are directly related to geometry assurance and tolerancing (Schleich 2017). Hence, this step is essential at the beginning of tolerancing to define the targets to be achieved by all the subsequent tolerancing activities.
- **Tolerance specification:** A clear and unambiguous specification of tolerances in compliance with international GPS (Geometrical Product Specification) standards is used to describe how the physical proportions, i.e., features, of the individual part geometries are allowed to deviate from their ideal (Kumar *et al.* 2009; Armillotta 2013; American Society of Mechanical Engineers 2018). It includes choosing dimensional and geometrical tolerance types for all functionally relevant features and datum reference frames (Armillotta 2013).
- **Tolerance allocation:** In this activity, one specific value for each specified tolerance callout is allocated to define the amount of acceptable variation, corresponding to the size of the tolerance intervals and zones (Armillotta 2013). This step is mainly based on experience, standards, textbooks, heuristics, rules of thumb, or more sophisticated approaches (Singh *et al.* 2009b) and is directly linked to the necessary effort in product realization.
- **Tolerance analysis:** The aim is to check if the previously defined part tolerances can assure product quality (Shen *et al.* 2005). It uses variation simulation and sensitivity analysis techniques to study the product under variations and assess the probabilistic response with the aid of quality metrics (Morse *et al.* 2018).
- **Tolerance synthesis:** The boundaries between tolerance allocation and synthesis are blurred as their definitions are often similar in literature (Roth 2024). Nonetheless, it is reasonable to see tolerance synthesis as an additional activity following tolerance analysis aiming to refine either the pre-defined allocation or specification scheme to meet the tolerancing requirements (Schleich 2017), in the best case in an optimal way in contrast to the initial allocation, see Figure 1.

3. A systematic description scheme for tolerancing

Aiming to answer the underlying research question, the development of a consistent description scheme is aligned with the Design Science Research

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

methodology according to (Hevner *et al.* 2004; vom Brocke, Hevner & Maedche 2020). The framework defined therein consists of the following interacting elements:

- the *environment* describing the people, organizations, or technologies, thus, emphasizing the relevance of the research
- the *knowledge base* including foundations and methodologies, thus, fostering the rigor of the research
- and the *research*, including the development and evaluation

In the first step of the methodological approach, the *environment* was explored, see Section 1, highlighting the interdisciplinary nature of tolerancing. Accordingly, tolerancing is characterized by various, sometimes interacting approaches, which cannot be described uniformly, thus hindering meaningful communication in either research or industry. The research within this contribution focuses on tolerancing in product design but also addresses interactions with related activities in product realization and other domains.

In the second step, related works were analyzed, aiming to concretize the research area and to form the basis for the development process. The identified works in Section 2.1, which address the structuring of tolerancing, are the *knowledge base* in this project and represent feedback options and possible starting points for its future application. Moreover, the definition of the fundamental tolerancing activities in Section 2.2 was set as a first sufficient basis for the development process, even though the activities might be carried out in different product design stages by different people in different contexts of use following different approaches and strategies. So, for example, the tolerance analysis can be utilized in the concept design even if the geometry is not fixed yet (Goetz, Schleich & Wartzack 2018b), in the final product design stage already considering the actual manufacturing conditions through process parameters (Schleich & Wartzack 2013), or in product realization mapping the accumulation of variations from multiple manufacturing steps (Hong & Chang 2002).

In the third step, the *research* itself took place, considering the *environment* and *knowledge base*. In the first place, the heterogeneity of tolerancing covering various people with different backgrounds, not only but primarily in design, manufacturing, and inspection (Nickolaisen 1999), as well as steadily changing objectives, constraints (robustness, quality, costs, manufacturability, etc.), and scopes of action (part, assembly, product and process level) over the product development stages indicated that a consistent and holistic description of tolerancing requires multiple categories. Thus, their definition, including their interrelations, was the first step in developing the description scheme. Therefore, numerous workshops were conducted between the authors, who have different viewpoints on tolerancing. Taking into account the approaches from the state of the art, commonalities were identified from which the categories shown in Figure 2 were derived, starting from the predefined key tolerancing activities, see Section 2.2. Despite differences in design, implementation, and terms used, the approaches focused on in research and practice intersect in the way of supporting these activities – they use the same basic *methods*. There are different definitions and understandings of the term "method" in literature, in particular in product design, and a clear distinction between the terms "methodology" and "method" is not always made (Gericke, Eckert & Stacey 2017). Nonetheless, methods can be understood as "systematic

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

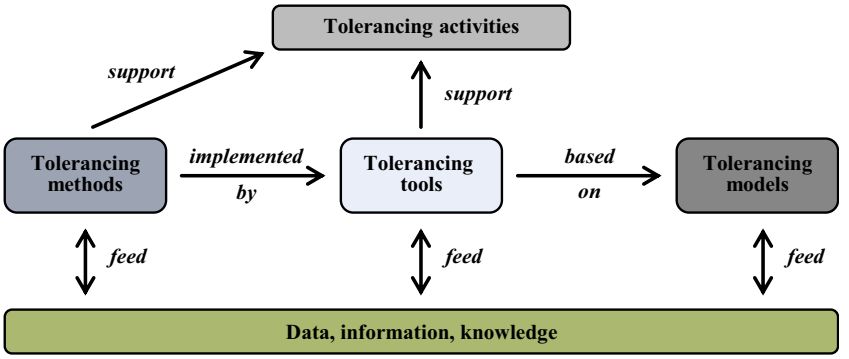


Figure 2. Mutual elements, namely activities, methods, tools, models and data, information, and knowledge, for describing existing tolerancing approaches.

procedure[s] with the intention to reach a specific goal" (Pahl *et al.* 2007) and "means to help designers achieve desired change as efficiently and effectively as possible" (Daalhuizen 2014). Hence, tolerancing methods can be referred to as a superset with different procedures as entities, answering the question of how to do something but differing in the level of prescription and detail (Gericke *et al.* 2017). Working aids in the form of software programs – be they scientific prototypes or proprietary software – are required to ensure the usability of tolerancing methods. In line with the definition given by Gericke *et al.* (2017), practical instruments supporting the application of tolerancing methods are named tolerancing *tools*. Their cores are both general and domain-specific tolerancing *models* representing reality in a simplified way through abstraction (Walter, Storch & Wartzack 2014). All these methods, tools, and models require input but also generate output in the form of raw, unorganized *data*, *information*, i.e., data in a context, and *knowledge*, i.e., interpreted information (Ahmed, Blessing & Wallace 1999). In summary, tolerancing can be defined by a small number of activities but in many different ways of implementation with different models, tools, methods, required data, information, and knowledge, as well as levels of detail.

This forms the basis for the proposed description scheme, whose details are introduced in the following, first focusing on product design in Section 3.1 and second extending its scope to manufacturing- and inspection-driven tolerancing in Section 3.2. Section 3.3 highlights the links to other domains before Section 3.4 summarizes the expected benefits. Following the Design Science Research methodology, these benefits were initially evaluated in Section 4.

3.1. Product design-driven tolerancing

To ensure the applicability of the description scheme, an illustrative visualization principle supporting communication and collaboration (Gebhardt, Beckmann, & Krause 2014; Gebhardt & Krause 2015; Kernbach & Svetina Nabergoj 2018) is utilized, following examples from product design with two-dimensional figurative flowcharts, e.g., the scenario funnel (Gausemeier, Fink & Schlake 1998) or the V-model for systems development (Clark 2009), and three-dimensional geometrical primitives, e.g., cubes for systems engineering (Hick, Bajzek & Faustmann 2019) or safety-oriented product design (Rajabalinejad 2018).

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

Considering the five common elements as different views, visualizing the scheme as a pyramid is an obvious choice. The lateral faces represent the activities, methods, tools, and models, while the fundamental connecting all four elements is data, information, and knowledge, representing the base of the pyramid, see Figure 3.

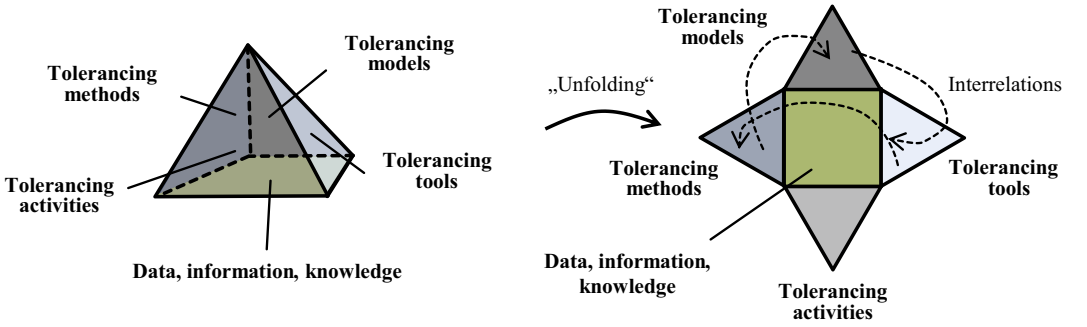


Figure 3. General structure of the proposed description scheme, visualized as a pyramid with its five mutual elements and interrelations.

The single elements of tolerancing or pyramid's faces intend to serve as general containers to collect individual entities to describe a given approach. For the main five activities, it makes sense to, e.g., arrange them according to their usual sequence, starting with basic activities essential for a complete tolerance scheme, including the definition of global tolerancing requirements, specification of part tolerances, and allocation of values, and continuing with optional activities of tolerance analysis and synthesis for making individual adjustments to tolerance specifications and their values, see Figure 4 (left).

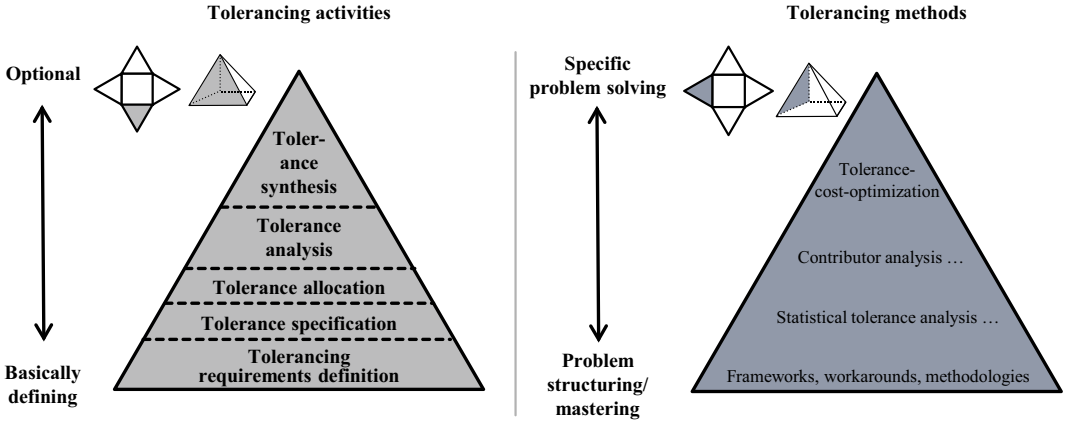


Figure 4. Allocation of the five fundamental tolerancing activities (left) and exemplary tolerancing methods (right) in the proposed pyramid description scheme.

In general, there is no absolute position where to place the entities on the respective triangular-shaped faces (near the bottom or the top). This basically differs, e.g., from the popular visualization of ISO GPS standards in a pyramid to

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

illustrate their hierarchy (ISO/TC 213 2011; Charpentier 2014). Nevertheless, it can be helpful to place fundamental entities near the base of the pyramid and the more specific ones near the top – even though the placement is just qualitative and up to the user's preferences. In contrast to the five activities, the entities of the other faces are not predefined. The diversity of methods, models, and tools and the non-standardized use of names severely complicate the definition of a complete list with fixed terms and definitions, even though it would foster clarity in the description of the approaches. Nonetheless, referring to exemplary entities in the following is sufficient for explaining the logical structure of the novel description scheme.

Figure 4 (right) shows the allocation of exemplary tolerancing methods, which range from basic problem structuring, e.g., using the Variation Management Framework (Howard *et al.* 2017), to specific problem-solving, e.g., tolerance-cost-optimization (Hallmann *et al.* 2020a). Focusing on the tools facet, assistance by computer and most likely specific computer-aided tolerancing software is common as it simplifies the application of tolerancing methods, alone or in combination, see Figure 5 (left). Examples include interactive tools, such as GD&T advisors ensuring tolerance specification consistency (Gaunet 2003; Anselmetti *et al.* 2010), commercial variation simulation software supporting tolerance analysis and automating the often challenging identification of tolerance chains (Glancy, Stoddard & Marvin 1999), as well as tools with a high level of automation, for example, to directly propose a tolerance specification based on machine learning (ML) (Cui *et al.* 2021). Tolerancing models range from tolerance representation and analysis models to knowledge models, with differentiation between mathematical or explicit, for instance, given in vectorial tolerance models (Cao *et al.* 2018), and numerical or implicit definitions, such as when using quantifiers (Dantan *et al.* 2005) or Finite Element models (Morse *et al.* 2018), see Figure 5 (right).

Regarding the bottom face of the pyramid, it is generally difficult to make an absolute distinction between data, information, and knowledge since it is relative, depending on how data is used (Ahmed *et al.* 1999). Despite critical opinions on its merit (Hubka & Eder 1996; Gericke *et al.* 2017), a differentiation can help to provide a first association with the type of the respective entity needed to feed the

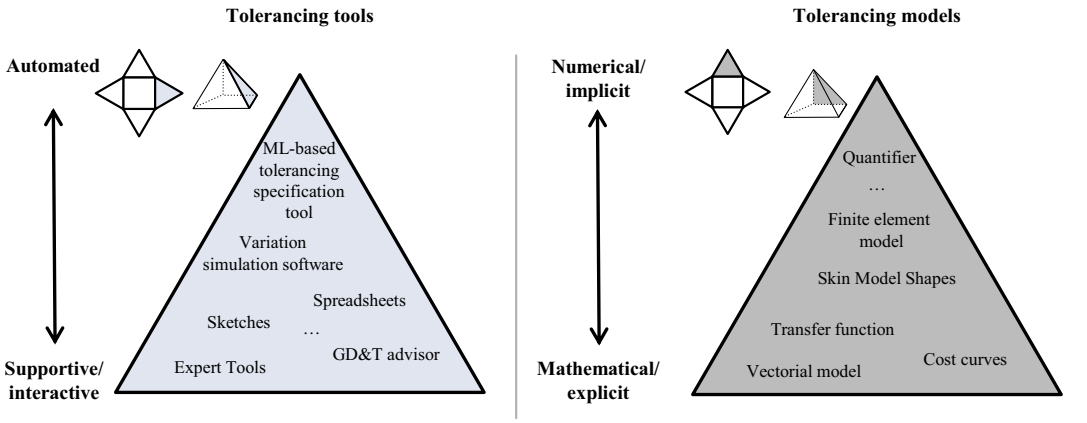


Figure 5. Allocation of exemplary tolerancing tools (left) and exemplary tolerancing models (right) in the proposed pyramid description scheme.

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

models, tools, and methods. Hence, the pyramid's base comprises raw data, such as measurement point cloud data of physical part entities (Wärmefjord *et al.* 2017), information, such as resulting part tolerance probability distributions for variation simulation (Hallmann *et al.* 2021b), and contextualized, respectively formalized knowledge, such as tolerancing rules for tolerance specification (Goetz & Schleich 2020), see Figure 6.

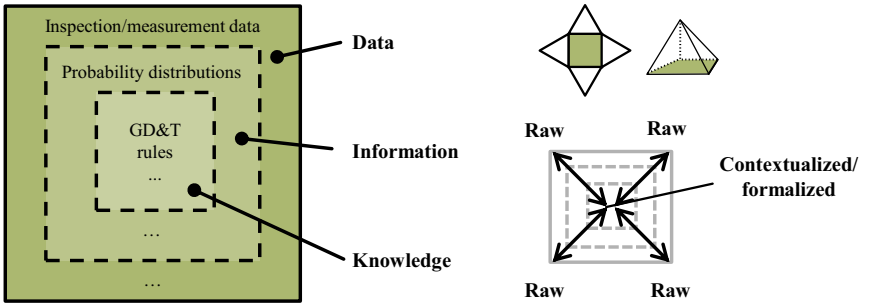


Figure 6. Allocation of exemplary tolerancing data, information, and knowledge in the proposed pyramid description.

Reviewing tolerancing approaches presented in literature shows that most approaches do not use just one single method, model, or tool. It is more of a set of entities to reach the specific goal of the respective activity. For example, sampling-based tolerance analysis, as it is called in Hallmann *et al.* (2021b), includes the use of a sampling method, such as Monte Carlo (MCS) or Latin Hypercube Sampling (LHS) (Hallmann *et al.* 2021b). From a more global viewpoint, this results in a large number of possible combinations or combination spaces of various methods, tools, and models, as well as the underlying data, information, and knowledge for the fundamental tolerancing activities – although the reason for the choice of a specific combination is often not further discussed. The interrelations and constraints, for example, given when using sampling and sensitivity analysis methods in combination (Saltelli 2008), are often only known by a few tolerancing experts. Formalizing this knowledge is the key to giving users clear guidance in finding an appropriate set of entities from all five faces. Starting from any pyramid face, it is then possible to systematically narrow down the solution space and find a suitable combination for a given context of use, see Figure 3 (right).

3.2. Linking product design-driven tolerancing to tolerancing in product realization

In product realization, the focus shifts from the virtual to the physical realm (Schleich 2017). For this reason, there is also a shift in responsibility. Manufacturing- and inspection-driven tolerancing activities become relevant to assure and verify the design-intended part and assembly quality. From the manufacturing perspective, they comprise activities to design the single- and multistage part and assembly manufacturing processes as well as their planning and organization. Inspection-driven tolerancing addresses all activities necessary to provide insights on deviations and variations (*The terms "deviation" and "variation" are often used*

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

as synonyms. To be more precise, "[t]he term geometrical deviation refers to the deviation from the nominal shape for one part or the mean-value deviation from the nominal shape for a set of parts. […] Geometrical variation is the variation of the deviation for different parts produced in the same process." (Wärmefjord, Hansen, & Söderberg 2023)), and achievable tolerances on both part and assembly levels. Despite the change of perspective, manufacturing- and inspection-driven tolerancing have strong overlaps in the entities of the five main elements of product-driven tolerancing. Figure 7 highlights examples of entities to illustrate the link between the different activities. Sampling-based variation simulation and contributor analysis, for instance, are general methods used for design, manufacturing, and inspection purposes (Söderberg *et al.* 2016; Morse *et al.* 2018). Thus, the same CAT tools, aiming for broad applicability, such as RD&T, can be applied for different activities (Söderberg *et al.* 2016). Besides, Skin Model Shapes (SMS) are one example of a general concept for the representation of variations that can be used along the product life cycle (Schleich *et al.* 2016; Hofmann, Gröger & Anwer 2022), whereas depending on the approach, a mapping between different representations may further be required, e.g., to enhance applicability by linking with domain-standard representations (Hallmann, Goetz, & Schleich 2019). Tolerancing data, information, and knowledge play an essential role since they directly connect all the individual activities (Schleich & Anwer 2021) defining their main in- and outputs. This applies not only to the use of product and GD&T information from the product design phase in all downstream activities but also to upstream feedback of, for instance, measurement data (Madrid *et al.* 2016), measurement uncertainty information (Kaufmann, Effenberger & Huber 2021), process capability information (Robles & Roy 2004), or knowledge about achievable manufacturing tolerances (Sauer *et al.* 2019) to design- and manufacturing-driven activities.

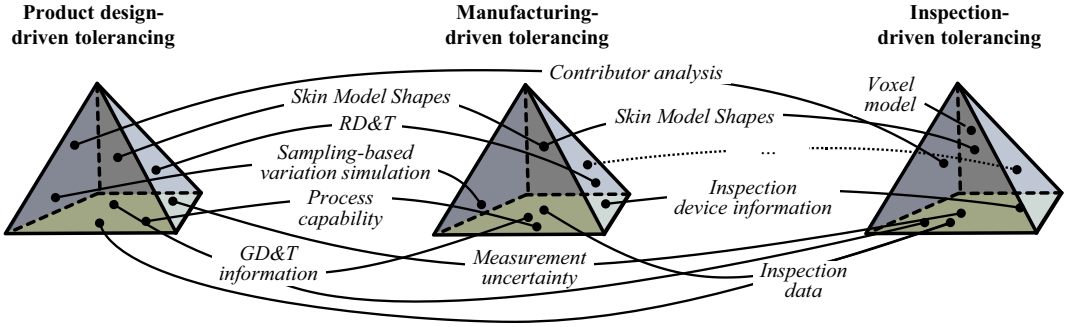


Figure 7. The five elements of the description scheme serve as links between product-, manufacturing- and inspection-driven tolerancing, illustrated by examples.

The result is a dense network of interrelations in which the individual entities connect the different tolerancing approaches. While this is supported, e.g., by the ISO GPS standards, the inherent principle of duality between specification and verification (ISO/TC 213 2011) corresponds to the differentiation of tolerance-related activities. Nevertheless, the meaning of interchangeability and interoperability of tolerancing data, information, and knowledge is undisputed, also emphasizing the role of neutral exchange formats to ensure the digital thread (Schleich *et al.* 2018; Schleich & Anwer 2021). Interoperability is, however, just as decisive for

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

methods, tools, and models to exploit synergy effects when individual entities are (re-)used across activities over the entire product life cycle.

In addition, there is a trend to shift individual activities from later stages to the design stage. Keeping the general vision of Concurrent Engineering alive (Smith 1997), integrated approaches frontload further activities to design-driven tolerancing, for example, machine and process selection or fixture layout design into tolerance synthesis (Rezaei Aderiani *et al.* 2021; Hallmann, Schleich & Wartzack 2021a). This makes particular sense for products where additional joining processes, such as welding or riveting, are used for assembly since they influence the final product design and quality far more than the individual part feature accuracies (Ding *et al.* 2005). Shifting of activities, however, has its limits since it can lead to conflicts, for example, when tolerance experts from product design largely define the production process and thus limit the scope of action for detailing the manufacturing process. Hence, the links through the elements and their entities are instrumental in bringing the disciplines closer together.

3.3. Linking tolerancing to further domains

Although tolerancing is a vast discipline of its own, it is crucial to keep in mind that it is just a small part of engineering, where "knowledge of the mathematical or physical sciences […] is applied" (Dugger 1993) (*referring to the original definition of the Accreditation Board for Engineering Technology (ABET)*). Hence, tolerancing is mainly based on general mathematics, statistics, and informatics principles, and it applies general methods, tools, and models. Consequently, all tolerancing approaches fall back on general entities and combine them in the context of use for tolerancing, forming tolerancing-specific models, methods, and tools. Metaphorically speaking, the three tolerancing pyramids from Figure 7 are surrounded by one or multiple further layers, depending on the level of abstraction of the observation. An explicit assignment of the entities to specific domains is not necessarily unambiguous. Figure 8 illustrates an outer pyramid as a totality of all elements not specific to tolerancing and shows exemplary interrelations. Tolerancing methods are, for example, often based on traditional methods from statistics, such as advanced sampling methods for tolerance analysis (Bacharoudis, Popov & Ratchev 2021), analysis of variance (ANOVA) to perform tolerance synthesis (Kusiak & Feng 1995), and fuzzy logic for estimation of tolerancing costs in tolerance synthesis (Dupinet, Balazinski & Czogala 1996), or machine learning and deep learning methods developed in computer science, such as for the generation of shapes with systematic and random variations (Qie, Schleich & Anwer 2023). In some cases, this requires a thorough adaptation of these traditional methods or the adaptation and mapping of tolerance-related data and information, such as distribution functions. A large number of tolerancing models are also based on general representations, such as Finite Element models (Franz, Schleich & Wartzack 2021), ontologies (Qin *et al.* 2018), or voxel models (Moroni, Petrò & Polini 2017). Speaking of tools, it becomes apparent that the presented approaches often use general software, such as spreadsheet software (Gerth 1994) or graph editors (Goetz *et al.* 2021b), as well as engineering tools, such as CAD (Goetz *et al.* 2018a) and Finite Element Analysis (FEA) software for structural analysis (Corrado & Polini 2018), as aids to achieve the goals in the respective activities. All these examples are just a few of many to illustrate the basic concept

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

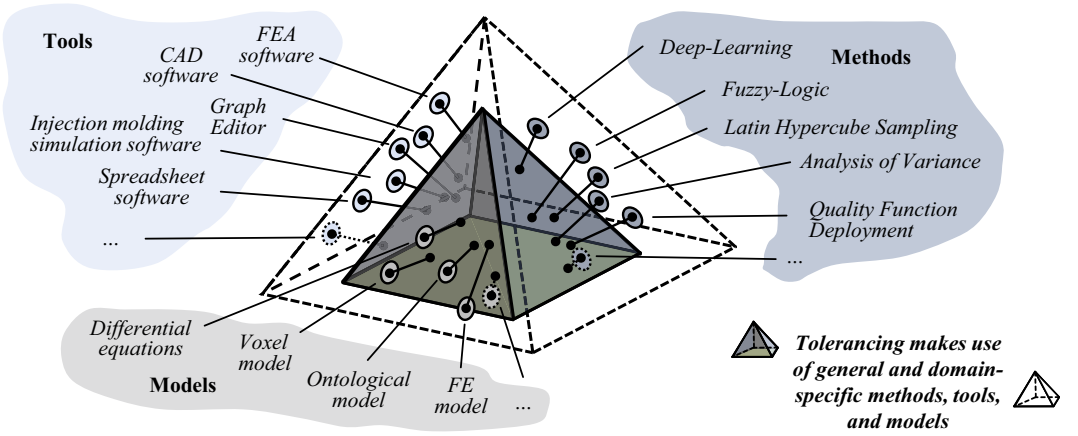


Figure 8. Models, methods, and tools link tolerancing to other engineering domains and disciplines, illustrated by examples.

that tolerancing may benefit from engineering approaches, e.g., by expanding the area of application or increasing efficiency.

Since product design-driven tolerancing is a part of product design, there are particularly strong links to design methods, models, and tools used in other activities. Manufacturing-driven tolerancing, in contrast, tends to rely on entities used to design the process, such as manufacturing simulation software (Schleich & Wartzack 2013). Characterizing the tolerancing approaches with the proposed elements and drawing interrelations between the entities on the tolerancing domain level, but also on the product design or higher levels, establishes a direct relationship to similar basic elements, e.g., sampling strategies or ontological models. In this way, similarities between tolerancing approaches, which at first glance appear very different, can be revealed. This helps less-experienced engineers get started and facilitates embedding the approaches in established product development processes. Moreover, it expands the view to identify overlaps with approaches outside the tolerancing domain.

3.4. Expected benefits of the proposed description scheme

The previous sections focused on the general structure of the description scheme, its implementation for product design-driven tolerancing, and its link to manufacturing- and inspection-driven tolerancing, see Section 3.2, as well as to other domains, see Section 3.3. While some expected benefits of the novel description scheme have already been mentioned in the individual passages, this section aims to consolidate the expected benefits that have driven the scheme's development.

The authors expect the main merit of the novel description scheme to serve as a practical instrument…

(1) … to classify and more clearly describe existing tolerancing approaches and procedures, either presented in scientific articles or applied in industry in a holistic way,

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

(2) … to facilitate the communication and collaboration between the different stakeholders involved in the variation management process from different domains with different backgrounds and wordings.

As a consequence, it aims to help…

(3) … to support the transfer of research methods into practice,
(4) … to facilitate the introduction and implementation of a thorough tolerancing process in practical applications,
(5) … to analyze existing tolerancing procedures implemented in the industry aiming to identify current shortcomings and potentials for improvement,
(6) … to identify current gaps and trends in literature, providing directions for future research works.

4. Application of the description scheme

For a first evaluation of the proposed scheme and its benefits, it is applied in two scenarios. First, it is employed for mapping tolerancing research works presented at an expert conference to demonstrate its merit in academia, see Section 4.1. Second, it is used for a consistent description of tolerancing activities in the development process of an exemplary chosen product showcasing the benefits for communication in practical applications, see Section 4.2.

4.1. Mapping of tolerancing research work

Aiming to overcome the problems associated with an unclear classification of the various tolerancing research approaches, see Section 1, the research works from the biennial CIRP Conference on Computer Aided Tolerancing are exemplarily mapped using the proposed description scheme. This conference represents the activities within the international tolerancing research community from different views and levels (Garaizar *et al.* 2014; Morse 2020), including the design, manufacturing, and inspection perspectives, see Figure 7.

The proceedings from the years 2020 and 2022 are the basis for this mapping. A total of 78 research papers were published at both conferences and initially classified according to the five fundamental tolerancing activities from a product design perspective, whereby papers solely addressing manufacturing-driven and inspection-driven tolerancing were not assigned to these design-driven tolerancing activities. While a complete presentation of all papers and their classification can be found in the table in the provided supplementary material, Figure 9 summarizes the classification according to the tolerancing activities. Thereby, multiple assignments are considered when one contribution addresses multiple activities, e.g., by providing an approach for both tolerance analysis and tolerance specification (Goetz, Lechner & Schleich 2022) or by unifying the design-driven tolerance synthesis with manufacturing- and inspection-driven activities (Dantan *et al.* 2022), thus demonstrating the strong linking along the product life cycle described in Section 3.2.

This classification indicates that, from a design perspective, the focus of the research presented at the conference was primarily on tolerance analysis, while the definition of tolerance requirements and tolerance allocation were barely considered. First, this reflects the scope of the conference. Second, it indicates that

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

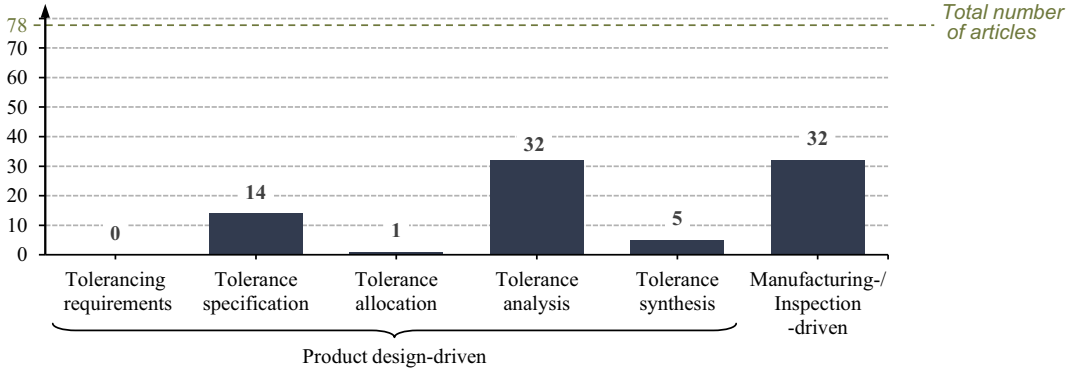


Figure 9. Illustration of the results from the classification of the research papers.

there is currently a lack of research in these areas. This may be because no acute need for research is apparent here since concrete, practical procedures, including realistic tolerance values, are of primary interest instead, for example, for the initial tolerance allocation. In addition, the close linking of tolerancing to other engineering domains, as discussed in Section 3.3, becomes evident, for example, for the tolerancing requirements definition, which adopts models, methods, and tools from requirements engineering and quality management, where extensive research driven by the emerging large language models takes place. Thus, taking into account a longer period of time, this classification fosters the identification of current research trends and focuses, as well as possible shifts to other areas.

This is primarily facilitated by the structured description scheme with predefined activities and elements. In contrast to titles or keywords, which can be arbitrarily defined by authors, this ensures a uniform description of tolerancing research works and thus helps to identify similarities. Within this scheme, the papers "An integrated resource allocation and tolerance allocation optimization: A statistical-based dimensional tolerancing" (Khezri *et al.* 2022) and "How to Consider Over-constrained Assemblies with Gaps in Tolerance-Cost Optimization?" (Hallmann, Schleich & Wartzack 2020b) similarly contribute to the tolerance synthesis activity, while the titles suggest otherwise due to the different methods, models, and tools as well as data, information, and knowledge used in areas with different backgrounds, see the table in the supplementary material. Thus, the proposed description scheme might facilitate communication between these areas.

Moreover, it supports bringing together different points of view and levels of detail, e.g., when proposing tolerance specification approaches utilizing knowledge from the ISO GPS system. While one publication addresses a method for the implementation of the GPS system in the practical tolerance specification process (Schuldt & Gröger 2022) from an organization or training perspective, the other proposes a method for the specification of highly interrupted collected features, such as textured surfaces (Fischer & Morse 2020). Although both pursue the goal of a standard-compliant tolerance specification, the two approaches are intended for different target groups, namely product developers and tolerance experts. While associated approaches are usually completely separate from each other, the

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

description via the five common elements enables the identification of intersections. It thus promotes possible cooperation across different levels in this case.

This also applies to approaches on similar levels, which, however, use different models for tolerance analysis, such as Skin Model Shapes (Shao, Liu & Ding 2020), polytope models (Teissandier, Delos & García 2020), or vector models (Schaechtl *et al.* 2020). If these approaches are assigned to similar tools, this fosters the combination of different models into a hybrid tolerance analysis model that unites the advantages of several models (Heling, Hallmann & Wartzack 2017). Besides, the proposed description scheme also facilitates the differentiation and potential integration on a detailed level. Thus, for example, the proposed open-source tolerance analysis tool based on Skin Model Shapes (Garcia *et al.* 2022) could be extended by a Skin Model Shape generation based on generative adversarial networks (Qie, Balaghi & Anwer 2022). Due to the usually inconsistent titles and keywords, this intersection also only becomes clear via the description scheme.

The application of the proposed scheme to the research work from the selected conferences highlights the need for a differentiated classification going beyond one purely based on just one category, e.g., the activity or just the title, which is not useful. This fosters an unambiguous description and classification of research work and enables easy structuring and clustering. Thus, it provides an overview of existing approaches with the corresponding methods, models, and tools as well as data, information, and knowledge for the different activities and facilitates differentiation and collaborations, ensuring scientific novelty.

4.2. Development of an electrified cross skate

While the previous section focused on the academic perspective, the proposed description scheme is subsequently applied in a practical tolerancing process scenario for a newly developed electrified cross skate. The scheme is used to describe the multiple, sometimes alternative, activities for ensuring and optimizing product quality in a structured manner. This is intended to simplify the communication between tolerance experts from different backgrounds and thus establish a procedure for targeted quality improvement that can be integrated into the development process.

The Technical Product Cross skates are roller skates with two wheels suitable for off-road use and enable skating like in cross-country skiing. During skating, upper and lower body muscles can be trained (Ross 2010). Hence, they offer a health-promoting micro-mobility solution to the so-called "last mile problem." The Institute of Engineering Design (KTfmk) of the Friedrich-Alexander-Universität Erlangen-Nürnberg took up this general idea and has developed an innovative, electrified version of a cross skate, called E-Cross Skate in the following (Kramer *et al.* 2023), see Figure 10. It is attached to the rider's foot by a binding, enabling its use with regular street shoes. The binding is mounted on four strain gauge load cells (two each at the front and rear), allowing driving power regulation of the electrically driven rear wheel. The front wheel is designed to be steerable to improve cornering and driveability, see the cross-section of the front wheel assembly in Figure 10 (right). For this purpose, a wheel hub steering system is suitable to meet the actuation and design space requirements. Hub-center steering is characterized by the steering pivot points located inside rather than outside or above the wheel hub as in conventional arrangements of steering mechanisms.

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

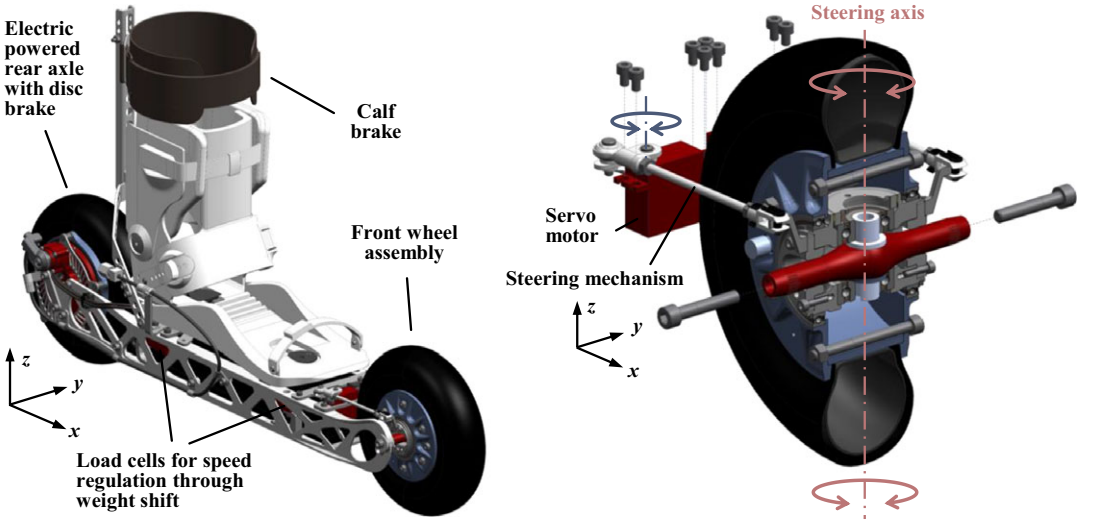


Figure 10. Overview of the CAD model of the E-Cross Skate (left) and its front wheel assembly (right) serving as the case study to emphasize the benefits of a tolerancing-supported product design.

Two servomotors redundantly actuate the steering system via steering rods depending on the inclination about the roll axis (x-axis in Figure 10), which is measured by an inclination sensor (gyroscope) on the controller board of each skate. The E-Cross Skate provides two driving modes. In the all-electric mode, the skates are aligned in parallel, and driving power is provided solely by the electric motors. In addition to purely electric driving, the E-Cross Skate features a new, patented hybrid driving mode (Rathert *et al.* 2022) in which the user moves in the skating technique as in cross-country skiing, supported by electric motors. The amount of auxiliary torque is determined based on the instantaneous distribution of the driver's weight between the two E-Cross Skates and among the individual load cells.

Tolerancing for E-Cross Skate The E-Cross Skate in Figure 10 shows the status of the prototype developed by the product design team. After verifying its functionality in physical tests, it had to be leveled up for series production by assuring product quality and robustness considering geometrical variations. It was the responsibility of the tolerancing team, formed by the dimensional management group of the institute of the authors, to improve the current status using different tolerancing approaches developed in research and established in industry but strongly differing in their focus. Hence, the following section does not aim to propose a chronological procedure with fixed activities. Instead, it illustrates the diversity of different activities with multiple people involved. At this time, the interdisciplinary team consisted of eight tolerancing experts (#1–#8), who are all part of the author team, with an experience level in tolerancing between 1.5 and 7.5 years (on average 4.1 years) and experience from industry in this field between 0 and 1.5 years (on average 0.3 years). Figures 11–12 and the following paragraphs provide a global overview of the different activities applied. More details on the activities can be found in their respective descriptions given in the supplementary material.

Initially, requirements relevant to tolerancing, such as Key Characteristics (KCs) or functions, were systematically identified and iteratively extended since

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

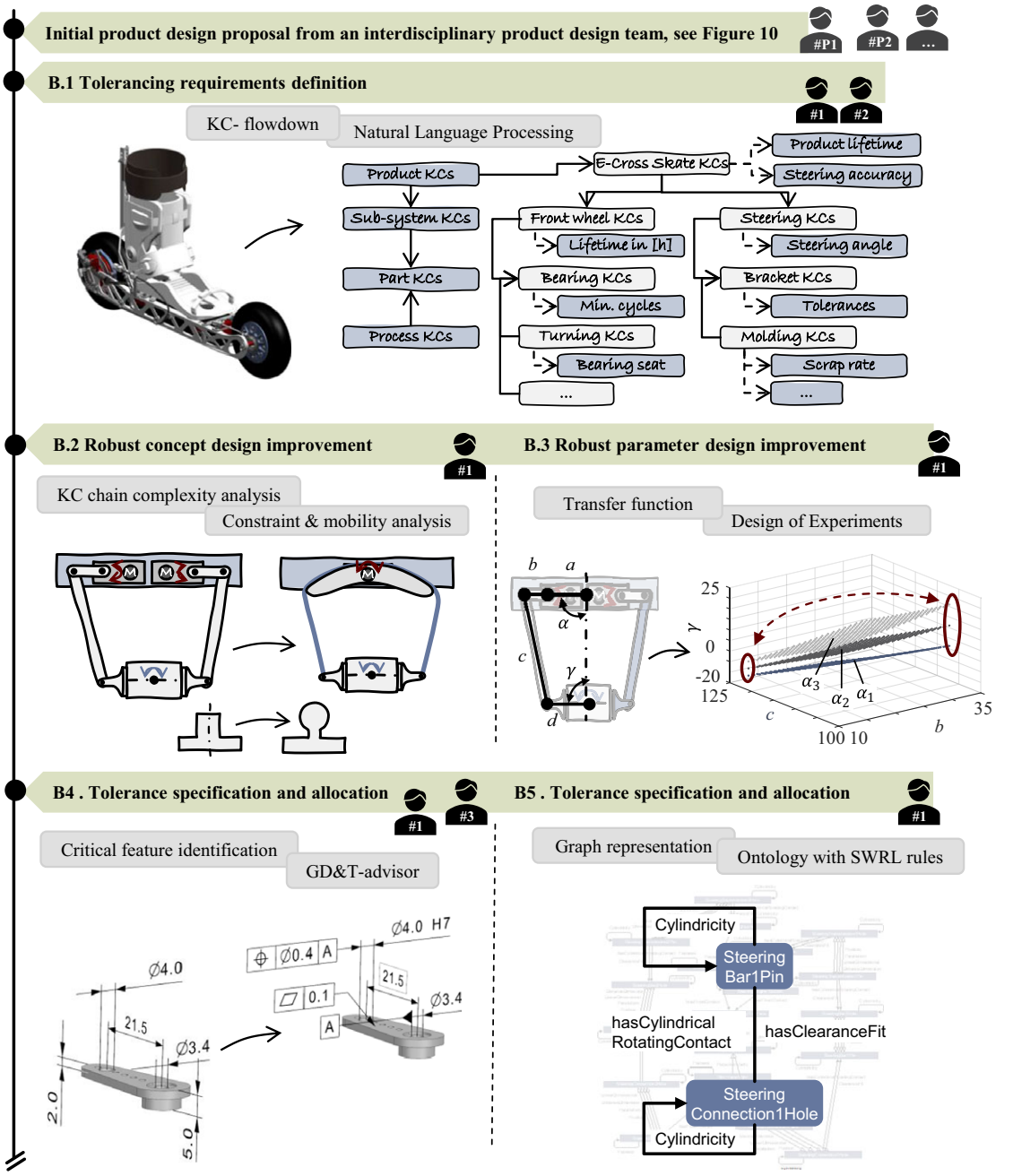


Figure 11. Overview of the different tolerancing activities carried out to assure the product quality of the E-Cross Skate (I/II).

the initial set of engineering requirements is usually incomplete, especially when product development progresses and new requirements need to be defined (Goetz *et al.* 2021a). By using the KC-flowdown method, the KCs were derived on different levels, from product KCs, e.g., the lifetime of the E-Cross Skate, over sub-systems,

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

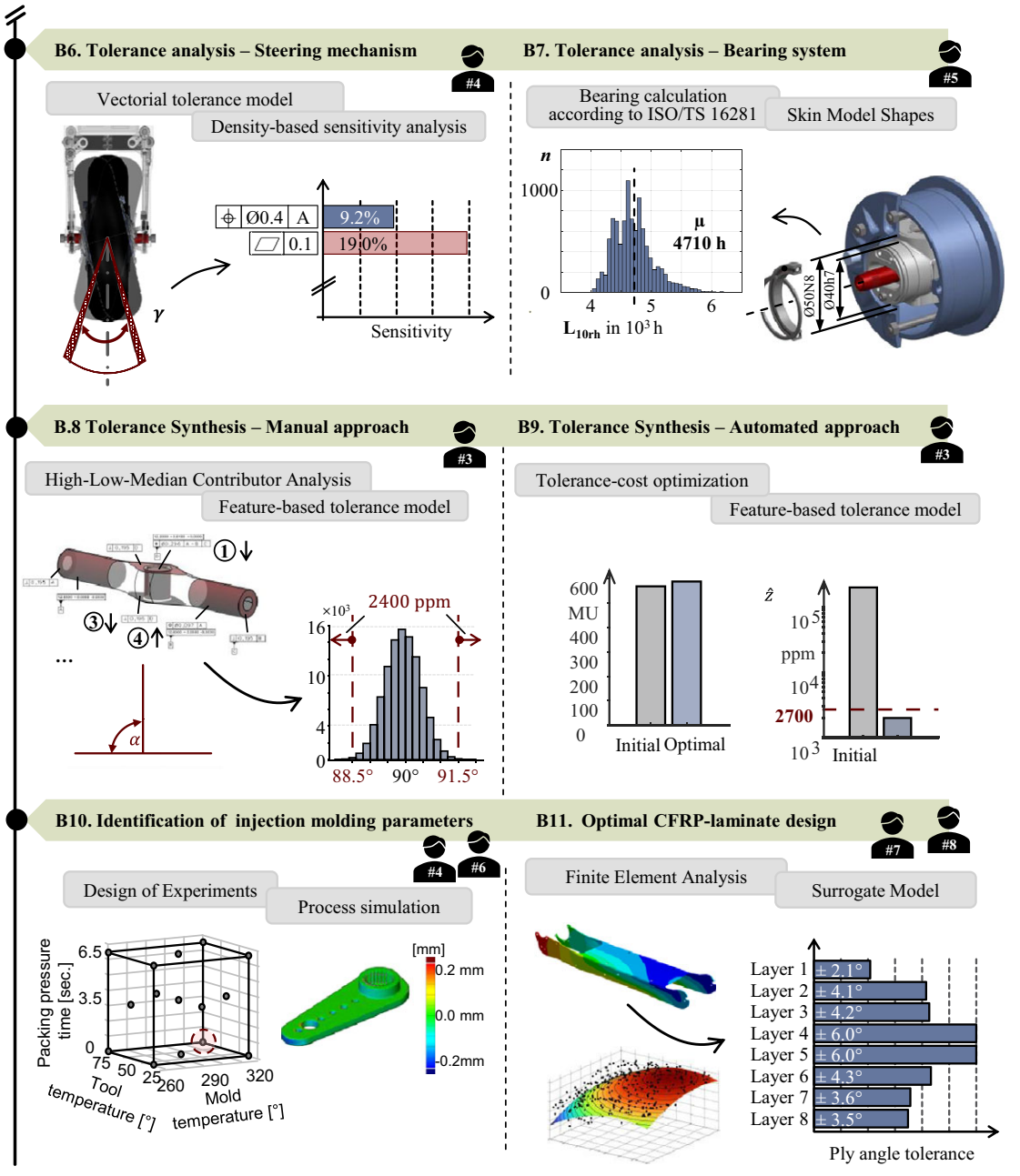


Figure 12. Overview of the different tolerancing activities carried out to assure the product quality of the E-Cross Skate (II/II).

e.g., the steering accuracy of the front wheel mechanism, and part KCs down to process KCs. Natural Language Processing (NLP) was used to extract the information from written product specifications (Goetz *et al.* 2021a), varying in detail and language. In this way, the tolerance experts #1 and #2 defined the *tolerancing*

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

requirements from documents and refined them in cooperation with the product design team, see Figure 11.

It is proven that enhancements of the concept and the parameter design in terms of robust design positively impact all subsequent tolerancing activities (Goetz, Roth & Schleich 2021c). Thus, expert #1 first compared two alternative concepts with either two motors with rods or one motor with cable drive for the front wheel steering and studied the overall robustness of the concepts, see Figure 11. Graphs transferred from initial concept sketches were used to automatically detect the KC chains, evaluate their complexity, and perform constraint and mobility analysis using screw theory to apply design concept improvements semi-automatically (Goetz *et al.* 2021b). The steering concept with two motors was identified as less robust than the cable solution. (*However, a change of concept typically causes a major redesign of the entire product and influences further requirements. Thus, although the cable drive concept was identified as a more robust solution, it was not chosen for the final design of the E-Cross Skate ready for series production, as the favored concept ensures redundancy.*) By using formalized Robust Design principles, a more *robust design concept*, e.g., through switching from revolute to ball joints, could be found, see Figure 11.

Subsequently, expert #1 searched for a robust combination of the parameters defining the steering mechanism rod lengths using transfer functions and Design of Experiments (DOE). Scatter plots for different configurations of the rod lengths and motor drive angles as inputs and the steering angle as output revealed the most *robust parameter design* of the mechanism, expressed through a lower sensitivity of the steering angle to changes in the drive angle, see Figure 11.

For subsequent *tolerance specification*, experts #1 and #3 followed a well-established manual approach. Step by step, they identified the individual contributions of each degree of freedom of the relevant features to the derived KC. Their variations are then limited through size, location, orientation, and form tolerances referring to current and internationally accepted standards on GD&T (ISO and ASME standards). The specification rules are implemented in most common CAD systems as GD&T-advisors, enabling the direct, model-based annotation of the GD&Ts as semantic product and manufacturing information (PMI), see Figure 11. For *tolerance allocation*, suggestions from handbooks and the ISO 2768 standard were used to complete the tolerance schemes for all parts of the front wheel assembly.

In addition to this manual approach, a knowledge-based approach for automated tolerance specification was used to verify the results. Therefore, the graph representation of the steering mechanism from the previous activities was extended from part to geometry element level, including contact definitions between the parts' faces and spatial relations among geometry elements within each part. In combination with tolerancing knowledge, formalized by an ontology including manifold Semantic Web Rule Language (SWRL) rules, proposals for *tolerance specification* and *allocation* for product functionality were automatically created and served as the basis to refine the manual tolerance allocation results, see Figure 11.

Subsequently, different experts focused on analyzing and improving the defined tolerance schemes. Expert #4 was responsible for *tolerance analysis* of the steering mechanism, i.e., studying if the proposals made were sufficient to limit the variation of the steering angle, defined as a geometrical, time-variant KC of the

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

system in motion (Walter, Sprügel & Wartzack 2013), see Figure 12. The time-variant interrelation between the geometrical tolerances and the steering angle as a function of time was represented by a vectorial tolerance model (Walter *et al.* 2013). Statistical variation simulation based on an LHS was used to create the inputs for density-based sensitivity analysis, to quantify the influence of the individual parts and their tolerances, and to give first hints of the most sensitive areas in the system (Schaechtl *et al.* 2020).

Bearing specialist and expert #5 focused on the tolerancing scheme of the bearings and their seats within the front wheel assembly since geometrical variations influence the internal bearing load distributions and thus bearing life (Aschenbrenner *et al.* 2020). For *tolerance analysis* of this non-geometrical KC, the geometry of the bearings and seats with their deviations was represented through Skin Model Shapes (Schleich *et al.* 2016), serving as the basis for the analysis of the bearing life (Kramer *et al.* 2024). The results indicated that a widening of the bearing seat diameter tolerances from g6 (shaft) and N7 (housing) to h7 and N8 does not decrease bearing life while fulfilling the mechanical performance requirements, see Figure 12.

These first hints from tolerance analysis bridge to the subsequent tolerance synthesis steps. The tolerance values pre-assigned in tolerance allocation were chosen for function while hoping that they are not cost-intensive. For this reason, expert #3 was responsible for *tolerance synthesis*, refining the initial tolerance schemes. First, the tolerance analysis software TCVisVA, based on a feature-based representation model, was used for variation simulation and sensitivity analysis. Taking the current tolerances into account, the predicted non-conformance rate of the camber of the E-Cross Skate's front wheel helped to evaluate if the tolerances are capable of assuring the required product quality, expressed by a maximum non-conformance rate in small parts-per-million (Hallmann *et al.* 2018). High-Low-Median (HLM) contributor analysis hints to the relevant tolerances to be manually tightened or widened. This helps to assure product quality and indirectly saves costs (Roth 2024). This manual, iterative adaptation and analysis loop was repeated until a satisfactory solution was achieved, see Figure 12.

Second, the cost aspect was included in the described *tolerance synthesis* loop. Tolerance-cost optimization was applied to minimize the sum of all tolerance-related manufacturing costs, expressed through relative tolerance-cost functions considering the feature type and geometry (Roth 2024). The method automates the manual search approach by metaheuristic optimization, where the tolerance values are balanced so they can assure product quality (here at maximum 2700 ppm) but with minimum costs to consider the economic impact of tolerancing already in the design phase (Hallmann *et al.* 2020a, 2021b), see Figure 12. (*The application of tolerance-cost optimization does not necessarily lead to lower costs compared to the initial status. It aims for the best combination of tolerances where the lowest costs can be achieved while ensuring product quality. Though the initial tolerances chosen for comparison in Figure 12 are less cost-intensive than the final solution, they are not capable of meeting the maximum non-conformance rate.*)

When switching from product design to product realization, tolerancing activities strongly focus on the manufacturing and inspection process, see Section 3.2. Two exemplary activities from manufacturing-driven tolerancing were applied to the E-Cross Skate. The process design for manufacturing one of the front

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

steering levers was focused on by the experts #4 and #6. They used virtual DOE jointly with a numerical process simulation tool to study the influence of the process on the final part geometry. Covering the relevant process variables for injection molding with their nominal values and variations mapping the real situation helped to virtually find a robust process window where the allocated tolerances can be achieved (Heling *et al.* 2019; Heling, Schleich & Wartzack 2020), see Figure 12.

In addition, the two experts, #7 and #8, in tolerancing of lightweight structures studied the potential of weight reduction when substituting the E-Cross Skate frame with a lightweight composite-fiber reinforced plastic (CFRP) part. Variation simulation was used to mimic the manufacturing process, covering ply angle variations on the laminate level and geometrical feature variations on the part level. Since the FEA-based process simulations are computationally expensive and time-consuming (Franz *et al.* 2021; Franz & Wartzack 2023), surrogate models were applied to be able to first analyze the resulting part deformations and their effect on the final product quality and costs and to second define the process variables in terms of ply angle tolerances in reasonable computing times, see Figure 12.

Evaluation of Using the Proposed Description Scheme In the first meeting, all experts proposed their approaches to be applied to the use case and discussed potential improvements. Although all involved participants were from the same institute, the chosen approaches differed according to the experts' roles and foci, with inconsistent descriptions and terminology hampering a common understanding. In a subsequent workshop, they used the proposed description scheme to structure their activities and described them in a standardized form using the five main elements and respective entities, see benefit (1) in Section 3.4. The individual feedback confirmed that it fostered communication and collaboration, see benefit (2) in Section 3.4. On the one hand, positive feedback within one activity with multiple stakeholders involved, such as in the tolerancing requirements definition or tolerance allocation and specification, see Figure 11, was reported. On the other hand, an enhanced cross-activity interaction was highlighted, facilitating working on parts shared within different activities, such as the tolerance values to the bearing seats assigned in tolerance analysis and considered within tolerance synthesis, see Figure 12. In addition, it simplified coordination between the tolerancing and the product design team, which was particularly necessary for early tolerancing activities, such as tolerancing requirements definition and robust concept and concept design improvement, see Figure 11. The scheme was perceived as beneficial for working in interdisciplinary teams with people with different backgrounds and levels of experience, as it fosters the accessibility of tolerancing topics for engineers even with little experience in tolerancing.

The structured descriptions presented in the supplementary material emphasize the high variability of the approaches expressed by the different entities chosen. While more qualitative approaches and manual expert knowledge are typically required in the early tolerancing phases, the ratio of simulation-driven methods increases in the later ones. Special applications, such as tolerancing of bearing assemblies or CFRP parts, see Figure 12, often require specific models, tools, and methods. Nevertheless, numerous similarities in the approaches are evident, some of which are not apparent at first glance. For instance, the parameter and process design approaches use DOE to study the influence of nominal parameters while

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

considering their variations. While the first focused on the robustness of the steering mechanism expressed by a transfer function in terms of the varying geometrical characteristics, the latter searched for robust process windows using numerical simulation tools to establish the relationship between varying process parameters and the final part geometry. Synergy effects can be exploited if these similarities are made recognizable via a common description. Hence, the developed scheme helped the tolerancing team to bring the approaches developed in research to practice and to define a continuous, common tolerancing process for a complex but still manageable product, see benefits (3)–(4) in Section 3.4.

5. Final discussion on the potentials and challenges

The proposed description scheme aims to overcome the drawbacks of a lacking overview of tolerancing approaches, which hinders the application of tolerancing and building on existing findings in research. Therefore, the complexity of tolerancing resulting from the interdisciplinarity and the varying levels of granularity is captured by a multidimensional description, distinguishing between the five elements of activity, method, model, and tool, as well as data, information, and knowledge. In contrast to a fixed universal tolerancing process, which is impossibly equally applicable to all scenarios and product development stages, an easy identification and combination of suitable methods, models, and tools to handle a specific tolerancing activity respecting the particular situation and the associated data, information, and knowledge is possible by making use of the novel scheme, see Section 3. Its three-dimensional visualization in pyramid form helps to emphasize the multidimensional character of tolerancing. As a practical aid, it can be unfolded and used as a worksheet on which the relationships can be directly highlighted, see Figure 3 (right). In addition, it may serve as a basis to create a computer-aided interactive advisor helping to navigate the multidimensional tolerancing maze, as presented, for example, in Stewart *et al.* (2020)) for finding suitable product development approaches for a given problem.

Moreover, the application of the proposed description scheme showed that it supports a clear classification of research work, see Section 4.1. It reduces ambiguities, resulting, for example, from misleading or differently understood titles and keywords, and makes research comparison and linking easier. This fosters the utilization of existing works and facilitates the identification of research trends and gaps. In alignment with the focus on tolerancing in product design, tolerancing can be broken down into five distinct key activities: tolerancing requirements definition, tolerance specification, tolerance allocation, tolerance analysis, and tolerance synthesis. While these five distinct entities ensure clarity, further concretization of the description of the activity, which can take place in different design stages, on different levels of detail, and with varying purposes, is done with the help of the other dimensions. Moreover, linking tolerancing in product design with manufacturing- and inspection-driven tolerancing enables focusing on essential aspects from other domains and thus reflecting the inherent interdisciplinarity, see Section 3.2. Depending on the preferred scope, other aspects of the product life cycle, such as the use phase, e.g., by incorporating wear or predictive maintenance aspects or the end-of-life phase, may also be linked in the future. The scheme's open structure offers the possibility for its continuous extension.

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

Furthermore, it becomes clear that tolerancing is directly connected to other domains and frequently uses well-established methods, tools, and models from mathematics, informatics, etc., see Section 3.3. The E-Cross Skate example in Section 4.2 further emphasized that virtual product development strongly affects tolerancing in product design and vice versa. This is particularly evident in Robust Design and when deriving the tolerance analysis models from the CAD model. This interdependence fosters the identification of a suitable tolerancing approach, considering the respective product development task and the associated constraints. However, a similar description of the product design approaches is necessary to create the link between the different approaches. Investigations on the transferability of the proposed multidimensional description to product development based on the classification of the extensive preliminary work carried out at the authors' institute in the field of virtual product development have shown a high potential for universal applicability. The introduction of the product design activity categories "Planning," "Generation of design," and "Design evaluation" (Cross 2000), as well as their mapping to tolerancing activities through the different elements in line with Figure 8, showed potential but has to be further studied in the future.

Despite the benefits, summarized at a glance in Section 3.4, applying the description scheme goes along with additional effort and is never completely unambiguous due to the inherent ambiguities of natural language, e.g., resulting from synonyms and homonyms. In the short run, the issues with ambiguities might be reduced by a common classification in heterogeneous teams, as is usual, e.g., in the product decision-making process. In the long run, a generally valid glossary, including distinct criteria and detailed explanations, with examples would be useful. This should be preferably created by major research associations in the field of tolerancing or by standardization committees. Thus, misconceptions could be avoided. Moreover, this could foster a clear distinction among all involved stakeholders between categories such as method and tool, which often seamlessly merge into one another. Nevertheless, the risk of incorrect classifications or descriptions remains in any case. However, the consequences are relatively small due to the multidimensional description. If, for example, a an SMS-based tolerance analysis approach was incorrectly assigned to the tolerance specification activity, it could still be found since it is assigned to the entity SMS in the model element. This ensures a basic resilience against misclassifications. Finally, the potential bias through human mistakes and the classification effort could be further reduced by future utilization of artificial intelligence. For example, ML based on already classified tolerancing approaches could support the classification of new works, or clustering algorithms might support the grouping of similar elements within the pyramid description scheme. Thereby, NLP could, on the one hand, facilitate this process and, on the other hand, support the interaction with the scheme, making it easily accessible. Therefore, a sufficient data basis considering different viewpoints and going beyond the classification described in the supplementary material would be beneficial. Additionally, it would enable an evaluation of the proposed approach and the identification of further potential for improvement. The description scheme could also be used with mandatory standardized metadata categories for the papers submitted to a tolerancing conference or a tolerancing-specific journal. Thus, clustering topics, sessions, and workshops and identifying research trends could be supported.

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

Moreover, extensive user studies with multiple stakeholders from various companies in different industrial sectors are necessary to verify the first qualitative findings on its usability in practical product development processes. The application of the scheme described in Section 4.2 serves only as an initial evaluation without claiming complete general validity and objectivity since the example of the E-Cross Skate is comparatively small and the people involved had similar backgrounds and levels of expertise. When applying the description scheme within larger, more heterogeneous companies, it can be assumed that the flexibility and extensibility addressed in Section 3.2 and Section 3.3 will become particularly relevant. Even though a controversial discussion between the stakeholders involved is to be expected, the open description scheme enables universal applicability, in contrast to a fixed tolerancing process with predefined formats and software.

6. Conclusion and outlook

This article introduced a novel description scheme for clustering and finding intersections and similarities between tolerancing approaches applied within individual product design stages. The classification into five mutual elements, i.e., activity, method, model, tool, and associated data, information, and knowledge, serves as the basis of the description scheme, represented by a pyramid with the five elements as five interrelated faces. In addition to its use for product design-driven tolerancing, the scheme ensures a clear differentiation and linkage to other disciplines and domains, enabling a holistic description of tolerancing along the product life cycle from an engineering perspective. Its application to recent research works in the tolerancing community and its exemplary use as an aid for structuring the tolerancing process of a complex product illustrate its benefits as a practical instrument for research and practice. Further user studies on a larger scale are needed to evaluate and improve its general usability. In addition, its extension by NLP approaches can ensure consistency. The findings can finally be used to set up a public database as a supportive system in research and practice.

Supplementary material

The supplementary material for this article can be found at http://doi.org/10.1017/dsj.2025.6.

Acknowledgements

The authors gratefully acknowledge the continuous financial support of the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) and the Arbeitsgemeinschaft industrieller Forschungsvereinigungen (AIF) over many years. Their funding of numerous tolerancing research projects allows the institute to make significant contributions to the field of tolerancing research. We would also like to thank numerous software providers for supporting us with a wide range of software products, see supplementary material, facilitating our research activities, and linking them to industrial practice.

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

Author contributions

Stefan Goetz: Conceptualization; Formal analysis; Investigation; Methodology; Project management; Software; Supervision; Visualization; Writing – original draft preparation; Writing – review & editing. **Martin Roth:** Conceptualization; Formal analysis; Investigation; Methodology; Software; Visualization; Writing – original draft preparation; Writing – review & editing. **Paul Schaechtl:** Formal analysis; Investigation; Software; Visualization; Writing – original draft preparation; Writing – review & editing. **Vincent Kramer:** Formal analysis; Investigation; Software; Visualization; Writing – review & editing. **Christoph Bode:** Formal analysis; Investigation; Software; Visualization; Writing –review & editing. **Dennis Horber:** Formal analysis; Investigation; Software; Visualization; Writing – review & editing. **Michael Franz:** Formal analysis; Investigation; Software; Visualization; Writing – review & editing. **Stephan Freitag:** Formal analysis; Investigation; Software; Visualization; Writing – review & editing. **Jörg Miehling:** Resources; Investigation; Writing – review & editing. **Sandro Wartzack:** Project management; Funding acquisition; supervision; Resources; Writing – review & editing.

Financial support

This research received no specific grant from any funding agency, commercial or not-for-profit sectors.

Competing interest

The authors declare no potential conflict of interest.

References

Ahmed, S., **Blessing, L.** & **Wallace, K.** 1999 The relationships between data, information and knowledge based on a preliminary study of engineering designers. In *Volume 3: 11th International Conference on Design Theory and Methodology, American Society of Mechanical Engineers, Las Vegas, Nevada, USA*, pp. 121–130.

American Society of Mechanical Engineers. 2018 ASME Y14.5–2018: Dimensioning and Tolerancing, Standard.

Anselmetti, B., **Chavanne, R.**, **Yang, J.-X.** & **Anwer, N.** 2010 Quick gps: a new cat system for single-part tolerancing. *Computer-Aided Design* **42**(9), 768–780; doi:10.1016/j.cad.2010.04.006.

Anwer, N., **Cid, G.** & **Mathieu, L.** 2003 XML based tolerance management for integrated design. *International CIRP Design Seminar* **785**, 1–12.

Armillotta, A. 2013 A method for computer-aided specification of geometric tolerances. *Computer-Aided Design* **45**(12), 1604–1616; doi:10.1016/j.cad.2013.08.007.

Armillotta, A. 2020 Selection of parameters in cost-tolerance functions: review and approach. *The International Journal of Advanced Manufacturing Technology* **108**(1–2), 167–182; doi:10.1007/s00170-020-05400-z.

Aschenbrenner, A., **Schleich, B.**, **Tremmel, S.** & **Wartzack, S.** 2020 A variational simulation framework for the analysis of load distribution and radial displacement of cylindrical roller bearings. *Mechanism and Machine Theory* **147**, 103769; doi:10.1016/j.mechmachtheory.2019.103769.

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

Bacharoudis, K., **Popov, A.** & **Ratchev, S.** 2021 Application of advanced simulation methods for the tolerance analysis of mechanical assemblies. In S. Ratchev, ed., *Smart Technologies for Precision Assembly*, Vol. **620**, pp. 153–167. Springer International Publishing.

Ballu, A., **Dufaure, J.** & **Teissandier, D.** 2007 An adaptive tolerance model for collaborative design. In P. F. Cunha & P. G. Maropoulos, eds, *Digital Enterprise Technology*, pp. 233–240.Springer.

Ballu, A., **Falgarone, H.**, **Chevassus, N.** & **Mathieu, L.** 2006 A new design method based on functions and tolerance specifications for product modelling. *CIRP Annals* **55**(1), 139–142; doi:10.1016/S0007-8506(07)60384-9.

Blessing, L. T. & **Chakrabarti, A.** 2009 *DRM, a Design Research Methodology*. Springer.

Bohn, M. & **Hetsch, K.** 2013 *Toleranzmanagement im Automobilbau*. Hanser.

Bradley, H. D. & **Maropoulosa, P. G.** (1998), A relation-based product model for computer supported early design assessment. *Journal of Materials Processing Technology* **76**(1–3), 88–95; doi:10.1016/S0924-0136(97)00328-2.

Cao, Y., **Liu, T.** & **Yang, J.** 2018 A comprehensive review of tolerance analysis models. *The International Journal of Advanced Manufacturing Technology* **97**(5–8), 3055–3085; doi: 10.1007/s00170-018-1920-2.

Charpentier, F. 2014 *Leitfaden für die Anwendung der Normen zur geometrischen Produktspezifikation (GPS), Beuth*.

Clark, J. O. 2009 System of systems engineering and family of systems engineering from a standards, V-model, and Dual-V model perspective. In *2009 3rd Annual IEEE Systems Conference*, pp. 381–387. IEEE.

Corrado, A. & **Polini, W.** 2018 Fea integration in the tolerance analysis using skin model shapes. *Procedia CIRP* **75**, 285–290; doi:10.1016/j.procir.2018.04.055.

Costadoat, R., **Mathieu, L.** & **Falgarone, H.** 2011 Design method taking into account geometric variations management along the design process. In A. Bernard, ed., *Global Product Development*, pp. 159–166. Springer.

Costadoat, R., **Mathieu, L.**, **Falgarone, H.** & **Fricero, B.** 2009 Integrated design method based on geometric variations to generate specification since the beginning of the product life-cycle. In *Proceedings of the 11th CIRP Conference on Computer Aided Tolerancing, Annecy, France*.

Craig, M. 1996 dimensional management versus tolerance assignment. *Assembly Automation* **16**(2), 12–16; doi:10.1108/01445159610117645.

Creveling, C. M. 1997 *Tolerance Design: A Handbook for Developing Optimal Specifications*. Addison-Wesley.

Cross, N. 2000 *Engineering Design Methods: Strategies for Product Design*, 3rd edn. Wiley.

Cui, L.-j., **Sun, M.-y.**, **Cao, Y.-l.**, **Zhao, Q.-j.**, **Zeng, W.-h.** & **Guo, S.-r.** 2021 A novel tolerance geometric method based on machine learning. *Journal of Intelligent Manufacturing* **32**(3), 799–821; doi:10.1007/s10845-020-01706-7.

Daalhuizen, J. 2014 Method Usage in Design – How methods function as mental tools for designers. TU Delft. PhD thesis.

Dantan, J.-Y., **Anwer, N.** & **Mathieu, L.** 2003 Integrated Tolerancing Process for conceptual design. *CIRP Annals* **52**(1), 135–138; doi:10.1016/S0007-8506(07)60549-6.

Dantan, J.-Y., **Ballu, A.** & **Mathieu, L.** 2008a Geometrical product specifications – model for product life cycle. *Computer-Aided Design* **40**(4), 493–501; doi:10.1016/j.cad.2008.01.004.

Dantan, J.-Y., **Etienne, A.**, **Mohammadi, M.**, **Khezri, A.**, **Homri, L.**, **Tavakkoli-Moghaddam, R.** & **Siadat, A.** 2022 Modular cost model for tolerance allocation, process

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

selection and inspection planning. *Procedia CIRP* **114**, 1–6; doi:10.1016/j.procir.2022.10.001.

Dantan, J. Y., **Hassan, A.**, **Etienne, A.**, **Siadat, A.** & **Martin, P.** 2008b Information modeling for variation management during the product and manufacturing process design. *International Journal on Interactive Design and Manufacturing (IJIDeM)* **2**(2), 107–118; doi:10.1007/s12008-008-0040-x.

Dantan, J.-Y. & **Homri, L.** 2024 Overview of mathematical concepts for tolerance analysis. *Procedia CIRP* **129**, 73–78, doi:10.1016/j.procir.2024.10.014.

Dantan, J.-Y., **Mathieu, L.**, **Ballu, A.** & **Martin, P.** 2005 Tolerance synthesis: quantifier notion and virtual boundary. *Computer-Aided Design* **37**(2), 231–240; doi:10.1016/j.cad.2004.06.008.

Desrochers, A. & **Laperrière, L.** 2003 Framework proposal for a modular approach to tolerancing. In P. Bourdet & L. Mathieu, eds., *Geometric Product Specification and Verification: Integration of Functionality*, pp. 207–216. Springer.

Ding, Y., **Jin, J.**, **Ceglarek, D.** & **Shi, J.** 2005 Process-oriented tolerancing for multi-station assembly systems. *IIE Transactions* **37**(6), 493–508; doi:10.1080/07408170490507774.

Drake, P. J. 1999 *Dimensioning and Tolerancing Handbook*. McGraw-Hill.

Dufaure, J. & **Tessandier, D.** 2008 A tolerancing framework to support geometric specifications traceability. *The International Journal of Advanced Manufacturing Technology* **36**(9–10), 894–907; doi:10.1007/s00170-006-0900-0.

Dugger Jr, W. E. 1993 The relationship between technology, science, engineering, and mathematics. In *Annual Conference of the American Vocational Association, Nashville, TN*, pp. 1–17.

Dupinet, É., **Balazinski, M.** & **Czogala, E.** 1996 Tolerance allocation based on fuzzy logic and simulated annealing. *Journal of Intelligent Manufacturing* **7**(6), 487–497; doi:10.1007/BF00122838.

Eifler, T. & **Schleich, B.** 2021 A robust design research landscape - review on the importance of design research for achieving product robustness. In *Proceedings of the Design Society: International Conference on Engineering Design (ICED21)*, Cambridge University Press pp. 211–220; doi:10.1017/pds.2021.22.

Feng, S. C. & **Song, E. Y.** 2000 Information modeling of conceptual design integrated with process planning. In *Recent Advances in Design for Manufacture (DFM)*, pp. 123–130. American Society of Mechanical Engineers.

Fischer, B. & **Morse, E.** 2020 Defining and controlling variation of highly-interrupted collected features. *Procedia CIRP* **92**, 158–162; doi:10.1016/j.procir.2020.05.181.

Franz, M., **Schleich, B.** & **Wartzack, S.** 2021 Tolerance management during the design of composite structures considering variations in design parameters. *The International Journal of Advanced Manufacturing Technology* **113**(5–6), 1753–1770; doi:10.1007/s00170-020-06555-5.

Franz, M. & **Wartzack, S.** 2023 Tolerance optimization of patch parameters for locally reinforced composite structures. *Applied Composite Materials* **30**(4), 1353–1376; doi:10.1007/s10443-022-10072-x.

Garaizar, O. R., **Anwer, N.**, **Mathieu, L.** & **Qiao, L.** 2014 Exploring the proceedings of the computer aided tolerancing cirp conferences and seminars: a scientometric analysis. In *Proceedings of the 13th CIRP conference on Computer Aided Tolerancing, Hangzhou, China*.

Garcia, C. A. R., **Teissandier, D.**, **Anwer, N.**, **Delos, V.**, **Ledoux, Y.** & **Pierre, L.** 2022 An integrated open source cat based on skin model shapes. *Procedia CIRP* **114**, 135–140; doi:10.1016/j.procir.2022.10.020.

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

Gaunet, D. 2003 3D functional tolerancing & annotation: catia tools for geometrical product specification. In P. Bourdet & L. Mathieu (eds.), *Geometric Product Specification and Verification: Integration of Functionality*, pp. 25–33. Springer.

Gausemeier, J., **Fink, A.** & **Schlake, O.** 1998 Scenario management. *Technological Forecasting and Social Change* **59**(2), 111–130; doi:10.1016/S0040-1625(97)00166-2.

Gebhardt, N., **Beckmann, G.** & **Krause, D.** 2014 Visual representation for developing modular product families – literature review and use in practice. In *International Design Conference (DESIGN 2014), Dubrovnik, Croatia*, The Design Society pp. 183–192.

Gebhardt, N. & **Krause, D.** 2015 On the development of visualisation concepts as tools in product design. In *DS 80–5 Proceedings of the 20th International Conference on Engineering Design (ICED 15) Vol 5: Design Methods, Milan, Italy*, The Design Society pp. 205–214.

Gericke, K., **Eckert, C.** & **Stacey, M.** 2017 What do we need to say about a design method?. In *DS 87–7 Proceedings of the 21st International Conference on Engineering Design (ICED 17) Vol 7: Design Theory and Research Methodology, Vancouver, Canada*, The Design Society pp. 101–110.

Germer, C. & **Franke, H.-J.** 2003 Interdisziplinäres toleranzmanagement. In *14th Symposium "Design for X", Neukirchen, Germany*, The Design Society pp. 49–58.

Gerth, R. J. 1994 A spreadsheet approach to minimum cost tolerancing for rocket engines. *Computers & Industrial Engineering* **27**(1–4), 549–552; doi:10.1016/0360-8352(94)90356-5.

Glancy, C. G., **Stoddard, J.** & **Marvin, L.** 1999 Automating the tolerancing process. In P. J. Drake, ed., *Dimensioning and Tolerancing Handbook*, pp.15-1–15-15. McGraw-Hill.

Goetz, S. 2022 Frühzeitiges konstruktionsbegleitendes Toleranzmanagement. *Friedrich-Alexander-Universität Erlangen-Nürnberg. PhD thesis. FAU Studienaus dem Maschinenbau, Nr.* 409; doi:10.25593/978-3-96147-594-0.

Goetz, S., **Horber, D.**, **Schleich, B.** & **Wartzack, S.** 2021a Simultaneous definition of key characteristics in order to facilitate robust design in early product development stages. *Proceedings of the Design Society* **1**, 2691–2700; doi:10.1017/pds.2021.530.

Goetz, S., **Kirchner, P.**, **Schleich, B.** & **Wartzack, S.** 2021b Integrated approach enabling robust and tolerance design in product concept development. *Design Science* **7**, e14; doi:10.1017/dsj.2021.13.

Goetz, S., **Lechner, T.** & **Schleich, B.** 2022 Computer-aided tolerance specification of preliminary designs based on variation analysis. *Procedia CIRP* **114**, 203–208; doi:10.1016/j.procir.2022.10.028.

Goetz, S., **Roth, M.** & **Schleich, B.** 2021c Early robust design—Its effect on parameter and tolerance optimization. *Applied Sciences* **11**(20), 9407; doi:10.3390/app11209407.

Goetz, S. & **Schleich, B.** 2020 Ontology-based representation of tolerancing and design knowledge for an automated tolerance specification of product concepts. *Procedia CIRP* **92**, 194–199; doi:10.1016/j.procir.2020.03.128.

Goetz, S., **Schleich, B.** & **Wartzack, S.** 2018a CAD-based tolerance analysis in preliminary design stages enabling early tolerance evaluation. In *Volume 2: Advanced Manufacturing*, pp. V002T02A106. American Society of Mechanical Engineers.

Goetz, S., **Schleich, B.** & **Wartzack, S.** 2018b A new approach to first tolerance evaluations in the conceptual design stage based on tolerance graphs. *Procedia CIRP* **75**, 167–172; doi:10.1016/j.procir.2018.04.030.

Goetz, S., **Schleich, B.** & **Wartzack, S.** 2020 Integration of robust and tolerance design in early stages of the product development process. *Research in Engineering Design* **31**(2), 157–173; doi:10.1007/s00163-019-00328-2.

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

Graves, S. & **Bisgaard, S.** 1999 Integrated tolerance management. In F. Van Houten & H. Kals, eds, *Global Consistency of Tolerances*, pp. 385–394. Springer.

Hallmann, M., **Goetz, S.** & **Schleich, B.** 2019 Mapping of GD&T information and PMI between 3D product models in the STEP and STL format. *Computer-Aided Design* **115**, 293–306; doi:10.1016/j.cad.2019.06.006.

Hallmann, M., **Schleich, B.**, **Heling, B.**, **Aschenbrenner, A.** & **Wartzack, S.** 2018 Comparison of different methods for scrap rate estimation in sampling-based tolerance-cost-optimization. *Procedia CIRP* **75**, 51–56; doi:10.1016/j.procir.2018.01.005.

Hallmann, M., **Schleich, B.** & **Wartzack, S.** 2020a From tolerance allocation to tolerance-cost optimization: a comprehensive literature review. *The International Journal of Advanced Manufacturing Technology* **107**(11–12), 4859–4912; doi:10.1007/s00170-020-05254-5.

Hallmann, M., **Schleich, B.** & **Wartzack, S.** 2020b How to consider over-constrained assemblies with gaps in tolerance-cost optimization?. *Procedia CIRP* **92**, 88–93; doi: 10.1016/j.procir.2020.05.168.

Hallmann, M., **Schleich, B.** & **Wartzack, S.** 2021a Process and machine selection in sampling-based tolerance-cost optimisation for dimensional tolerancin. *International Journal of Production Research* **60**(17), 5201–5216; doi:10.1080/00207543.2021.1951867.

Hallmann, M., **Schleich, B.** & **Wartzack, S.** 2021b Sampling-based tolerance analysis: the key to establish tolerance-cost optimization in the product development process. *Procedia CIRP* **100**, 560–565; doi:10.1016/j.procir.2021.05.123.

Heling, B., **Hallmann, M.** & **Wartzack, S.** 2017 Hybrid tolerance representation of systems in motion. *Procedia CIRP* **60**, 50–55; doi:10.1016/j.procir.2017.02.048.

Heling, B., **Oberleiter, T.**, **Rohrmoser, A.**, **Kiener, C.**, **Schleich, B.**, **Hagenah, H.**, **Merklein, M.**, **Willner, K.** & **Wartzack, S.** 2019 A concept for process-oriented interdisciplinary tolerance management considering production-specific deviations. *Proceedings of the Design Society: International Conference on Engineering Design* **1**(1), 3441–3450; doi:10.1017/dsi.2019.351.

Heling, B., **Schleich, B.** & **Wartzack, S.** 2020 An approach for determining the influence of manufacturing process parameters on product quality characteristics. *Procedia CIRP* **92**, 212–217; doi:10.1016/j.procir.2020.05.179.

Henzold, G. 2021 *Geometrical Dimensioning and Tolerancing for Design, Manufacturing and Inspection*, 3rd edn. Elsevier.

Hevner, A. R., **March, S. T.**, **Park, J.** & **Ram, S.** 2004 Design science in information systems research. *MIS Quarterly* **28**, 75–105; doi:10.2307/25148625.

Hick, H., **Bajzek, M.** & **Faustmann, C.** 2019 Definition of a system model for model-based development. *SN Applied Sciences* **1**(9), 1074; doi:10.1007/s42452-019-1069-0.

Hofmann, R., **Gröger, S.** & **Anwer, N.** 2022 Evaluation of relations and accumulations of geometrical deviations in multi-stage manufacturing based on skin model shapes. *Procedia CIRP* **114**, 147–152; doi:10.1016/j.procir.2022.10.038.

Hong, Y. S. & **Chang, T. C.** 2002 A comprehensive review of tolerancing research. *International Journal of Production Research* **40**(11), 2425–2459; doi:10.1080/00207540210128242.

Horber, D., **Goetz, S.**, **Schleich, B.** & **Wartzack, S.** 2022 A model-based approach for integrated variation management. In *Volume 2B: Advanced Manufacturing*, pp. V02BT02A067. American Society of Mechanical Engineers.

Howard, T. J., **Eifler, T.**, **Pedersen, S. N.**, **Göhler, S. M.**, **Boorla, S. M.** & **Christensen, M. E.** 2017 The variation management framework (vmf): a unifying graphical representation of robust design. *Quality Engineering* **29**(4), 563–572; doi:10.1080/08982112.2016.1272121.

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

Hu, J. & **Peng, Y.** 2007 Tolerance modelling and robust design for concurrent engineering. *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science* **221**, 455–465; doi:10.1243/0954406JMES438.

Hubka, V. & **Eder, W. E.** 1996 *Design Science. Introduction to the Needs, Scope and Organization of Engineering Design Knowledge*. Springer.

Islam, M. N. 2009 A dimensioning and tolerancing methodology for concurrent engineering applications I: problem representation. *The International Journal of Advanced Manufacturing Technology* **42**(9–10), 910–921; doi:10.1007/s00170-008-1649-4.

ISO/TC 213 2011 ISO 8015:2011: Geometrical product specifications (GPS) - Fundamentals - Concepts, principles and rules, Standard.

ISO/TC 213 2014 ISO 14638:2015: Geometrical product specifications (GPS) - matrix model, Standard.

Johannesson, H. & **Söderberg, R.** 2000 Structure and matrix models for tolerance analysis from configuration to detail design. *Research in Engineering Design* **12**(2), 112–125; doi:10.1007/s001630050027.

Karmakar, S. & **Maiti, J.** 2012 A review on dimensional tolerance synthesis: paradigm shift from product to process. *Assembly Automation* **32**(4), 373–388; doi:10.1108/01445151211262438.

Kaufmann, M., **Effenberger, I.** & **Huber, M. F.** 2021 Measurement uncertainty assessment for virtual assembly. *Journal of Sensors and Sensor Systems* **10**(1), 101–108; doi:10.5194/jsss-10-101-2021.

Kernbach, S. & **Svetina Nabergoj, A.** 2018 Visual design thinking: understanding the role of knowledge visualization in the design thinking process. In *2018 22nd International Conference Information Visualisation (IV)*, pp. 362–367. IEEE.

Khezri, A., **Homri, L.**, **Etienne, A.** & **Dantan, J.-Y.** 2022. An integrated resource allocation and tolerance allocation optimization: a statistical-based dimensional tolerancing. *Procedia CIRP* **114**, 88–93; doi:10.1016/j.procir.2022.10.012.

Kramer, V., **Atalay, O.**, **Rüth, L.**, **Wingertszahn, P.**, **Bartz, M.**, **Goetz, S.**, **Schleich, B.**, **Koch, O.** & **Wartzack, S.** 2024. Method for the consideration of statistically distributed geometrical deviations in the design of cylindrical roller bearing arrangements. *Forschung im Ingenieurwesen* **88**(1), 30; doi:10.1007/s10010-024-00749-z.

Kramer, V., **Wirsching, S.**, **Wartzack, S.** & **Miehling, J.** 2023 Electrified cross skate as a micromobility solution. *Konstruktion* **75**(9), 59–66; doi:10.37544/0720-5953-2023-09-59.

Krogstie, L. & **Martinsen, K.** 2012 Closed loop tolerance engineering – a relational model connecting activities of product development. *Procedia CIRP* **3**, 519–524; doi:10.1016/j.procir.2012.07.089.

Krogstie, L., **Martinsen, K.** & **Andersen, B.** 2014 Approaching the devil in the details; A survey for improving tolerance engineering practice. *Procedia CIRP* **17**, 230–235; doi:10.1016/j.procir.2014.01.054.

Krogstie, L., **Walter, M. S.**, **Wartzack, S.** & **Martinsen, K.** 2015 Towards a more comprehensive understanding of tolerance engineering research importance. *Procedia CIRP* **27**, 29–34; doi:10.1016/j.procir.2015.04.039.

Kumar, S., **Soundararajan, V.** & **Alagumurthi, N.** 2009 Review of tolerance analysis, allocation and constraints in manufacturing. *Journal for Manufacturing Science and Production* **10**(1), 1–16; doi:10.1515/IJMSP.2009.10.1.1.

Kusiak, A. & **Feng, C.-X.** 1995 Deterministic tolerance synthesis: a comparative study. *Computer-Aided Design* **27**(10), 759–768; doi:10.1016/0010-4485(94)00028-C.

Leidich, E., **Ebermann, M.** & **Gröger, S.** 2014 Integration der GPS in den methodischen Konstruktionsprozess nach VDI 2221. In R. Stelzer, ed., *ENTWERFEN ENTWICKELN*

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

ERLEBEN 2014: Beiträge Zur Virtuellen Produktentwicklung Und Konstruktionstechnik, pp. 383–396. TUD Press.

Madrid, J., **Vallhagen, J.**, **Söderberg, R.** & **Wärmefjord, K.** 2016 Enabling reuse of inspection data to support robust design: a case in the Aerospace Industry. *Procedia CIRP* **43**, 41–46; doi:10.1016/j.procir.2016.02.137.

Maltauro, M., **Hofmann, R.**, **Concheri, G.**, **Meneghello, R.** & **Gröger, S.** 2024a Content evolution in ISO GPS documents in product development. *Procedia CIRP* **129**, 55–60; doi:10.1016/j.procir.2024.10.011.

Maltauro, M., **Meneghello, R.** & **Concheri, G.** 2024b Tolerance specifications management integrated into the product development cycle. *Machines* **12**(2), 147; doi:10.3390/machines12020147.

Marziale, M. & **Polini, W.** 2011 Review of variational models for tolerance analysis of an assembly. *Proceedings of the Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture* **225**(3), 305–318; doi:10.1177/2041297510394107.

Mathieu, L. & **Ballu, A.** 2007 A model for a coherent and complete tolerancing process. In J. K. Davidson, ed., *Models for Computer Aided Tolerancing in Design and Manufacturing*, pp. 35–44. Springer.

Mölzer, B. & **Strobelt, M.** 2013 *Toleranzmanagement in der Fahrzeugentwicklung: Eine präventive Qualitätsmethode*, pp. 24–29. Porsche Engineering Magazin.

Moroni, G., **Petrò, S.** & **Polini, W.** 2017 Geometrical product specification and verification in additive manufacturing. *CIRP Annals* **66**(1), 157–160; doi:10.1016/j.cirp.2017.04.043.

Morse, E. 2020 Opportunities and constraints in the standardization of geometrical product specification. *Procedia CIRP* **92**, 1–2; doi:10.1016/j.procir.2020.07.001.

Morse, E., **Dantan, J.-Y.**, **Anwer, N.**, **Söderberg, R.**, **Moroni, G.**, **Qureshi, A.**, **Jiang, X.** & **Mathieu, L.** 2018 Tolerancing: managing uncertainty from conceptual design to final product. *CIRP Annals* **67**(2), 695–717; doi:10.1016/j.cirp.2018.05.009.

Ngoi, B. K. A. & **Ong, C. T.** 1998 Product and process dimensioning and tolerancing techniques. A state-of-the-art review. *The International Journal of Advanced Manufacturing Technology* **14**(12), 910–917; doi:10.1007/BF01179081.

Nickolaisen, R. H. 1999 Dimensional management. In P. J. Drake, ed., *Dimensioning and Tolerancing Handbook*, pp. 2–1–2–1. McGraw-Hill.

Nigam, S. D. & **Turner, J. U.** 1995 Review of statistical approaches to tolerance analysis. *Computer-Aided Design* **27**(1), 6–15; doi:10.1016/0010-4485(95)90748-5.

Pahl, G., **Beitz, W.**, **Feldhusen, J.**, **Grote, K.-H.**, **Wallace, K.** & **Blessing, L.**, eds 2007 *Engineering Design: A Systematic Approach*, 3rd edn. Springer.

Qie, Y., **Balaghi, M.** & **Anwer, N.** 2022 Wasserstein generative adversarial networks for form defects modeling. *Procedia CIRP* **114**, 7–12; doi:10.1016/j.procir.2022.10.002.

Qie, Y., **Schleich, B.** & **Anwer, N.** 2023 Generative adversarial networks and hessian locally linear embedding for geometric variations management in manufacturing. *Journal of Intelligent Manufacturing* **36**, 1033–1062 doi:10.1007/s10845-023-02284-0.

Qin, Y., **Qi, Q.**, **Lu, W.**, **Liu, X.**, **Scott, P. J.** & **Jiang, X.** 2018 A review of representation models of tolerance information. *The International Journal of Advanced Manufacturing Technology* **95**(5–8), 2193–2206; doi:10.1007/s00170-017-1352-4.

Rajabalinejad, M. 2018 Incorporation of safety into design by safety cube. *International Journal of Industrial and Manufacturing Engineering* **12**(3), 476–479.

Rathert, T. A., **Miehling, J.**, **Wartzack, S.**, **Biehler, M.**, **Kramer, V.**, **Duschl, S.** & **Kuhn, D.** 2022 Verfahren zum Betrieb eines Paares Skates, Patent DE102019212389B4, Germany.

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

Rezaei Aderiani, A., **Hallmann, M.**, **Wärmefjord, K.**, **Schleich, B.**, **Söderberg, R.** & **Wartzack, S.** 2021. Integrated tolerance and fixture layout design for compliant sheet metal assemblies. *Applied Sciences* **11**(4), 1646; doi:10.3390/app11041646.

Rhahli, Y., **Bosch-Mauchand, M.**, **Anselmetti, B.** & **Eynard, B.** (2010), A survey of tolerancing task integration in PLM. In *Proceedings of IDMME - Virtual Concept 2010, Bordeaux, France*, Springer pp. 1–6.

Robles, N. & **Roy, U.** 2004 Optimal tolerance allocation and process-sequence selection incorporating manufacturing capacities and quality issues. *Journal of Manufacturing Systems* **23**, 127–133 doi:10.1016/S0278-6125(05)00002-6.

Ross, J. A. 2010 Skiing and Snowboarding. In M. B. Werd & E. L. Knight, eds, *Athletic Footwear and Orthoses in Sports Medicine*, pp. 267–274. Springer.

Roth, M. 2024 Sampling-based tolerance-cost optimization: the key to optimal tolerance allocation. *Friedrich-Alexander-Universität Erlangen-Nürnberg*. PhD thesis. FAU Studien aus dem Maschinenbau, Nr 436; doi:10.25593/978-3-96147-720-3.

Roy, U., **Liu, C.** & **Woo, T.** 1991 Review of dimensioning and tolerancing: representation and processing. *Computer-Aided Design* **23**(7), 466–483; doi:10.1016/0010-4485(91)90045-X.

Roy, U., **Pramanik, N.**, **Sudarsan, R.**, **Sriram, R.** & **Lyons, K.** 2001 Function-to-form mapping: model, representation and applications in design synthesis. *Computer-Aided Design* **33**(10), 699–719; doi:10.1016/S0010-4485(00)00100-7.

Saltelli, A. 2008 *Global Sensitivity Analysis: The Primer*. John Wiley.

Sauer, C., **Heling, B.**, **Schmutzler, S.** & **Schleich, B.** 2019 A knowledge-based engineering workbench for automated tolerance specification. In *Proceedings of the ASME 2019 International Mechanical Engineering Congress and Exposition. Volume 2B: Advanced Manufacturing*, pp. V02BT02A062. American Society of Mechanical Engineers.

Schaechtl, P., **Hallmann, M.**, **Schleich, B.** & **Wartzack, S.** 2020 Tolerance analysis of additively manufactured non-assembly mechanisms considering joint clearance. *Procedia CIRP* **92**, 27–32; doi:10.1016/j.procir.2020.04.140.

Schleich, B. 2017 Skin model shapes: A new paradigm for the tolerance analysis and the geometrical variations modelling in mechanical engineering. *Friedrich-Alexander-Universität Erlangen-Nürnberg. PhD thesis. Fortschritt-Berichte VDI Reihe 1,Konstruktionstechnik, Maschinenelemente Nr*. 438; doi:10.51202/9783186438010.

Schleich, B. & **Anwer, N.** 2021 Tolerancing informatics: towards automatic tolerancing information processing in geometrical variations management. *Applied Sciences* **11**(1), 198; doi:10.3390/app11010198.

Schleich, B., **Anwer, N.**, **Mathieu, L.** & **Wartzack, S.** 2016 Status and prospects of skin model shapes for geometric variations management. *Procedia CIRP* **43**, 154–159; doi:10.1016/j.procir.2016.02.005.

Schleich, B., **Anwer, N.**, **Zhu, Z.**, **Qiao, L.**, **Mathieu, L.** & **Wartzack, S.** 2014 A Comparative Study on Tolerance Analysis Approaches. In *Proceedings of the 1st International Symposium on Robust Design, Copenhagen, Denmark*, pp. 29–39.

Schleich, B., **Wärmefjord, K.**, **Söderberg, R.** & **Wartzack, S.** 2018 Geometrical variations management 4.0: towards next generation geometry assurance. *Procedia CIRP* **75**, 3–10; doi:10.1016/j.procir.2018.04.078.

Schleich, B. & **Wartzack, S.** 2013 Process-oriented tolerancing - A discrete geometry framework. In *DS 75–5: Proceedings of the 19th International Conference on Engineering Design (ICED13) Design For Harmonies, Vol. 5: Design for X, Design to X*, The Design Society pp. 61–70. Seoul, Korea.

Schleich, B. & **Wartzack, S.** 2014 How can computer aided tolerancing support closed loop tolerance engineering?. *Procedia CIRP* **21**, 312–317; doi:10.1016/j.procir.2014.03.129.

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

Schuldt, J. & **Gröger, S.** 2022 The assessment of the iso gps system implementation with a gps maturity model. *Procedia CIRP* **114**, 197–202; doi:10.1016/j.procir.2022.10.027.

Shao, N., **Liu, J.** & **Ding, X.** 2020 Effects of contact behaviors on tolerance analysis of mechanism based on skin model shapes and a boundary element method. *Procedia CIRP* **92**, 15–20; doi:10.1016/j.procir.2020.05.173.

Shen, Z., **Ameta, G.**, **Shah, J. J.** & **Davidson, J. K.** 2005 A comparative study of tolerance analysis methods. *Journal of Computing and Information Science in Engineering* **5**(3), 247–256; doi:10.1115/1.1979509.

Shen, Z., **Shah, J. J.** & **Davidson, J. K.** 2008 Analysis neutral data structure for gd&t. *Journal of Intelligent Manufacturing* **19**(4), 455–472; doi:10.1007/s10845-008-0096-2.

Sigurdarson, N., **Eifler, T.** & **Ebro, M.** 2018 The applicability of cat tools in industry – boundaries and challenges in tolerance engineering practice observed in a medical device company. *Procedia CIRP* **75**, 261–266; doi:10.1016/j.procir.2018.04.066.

Singh, P. K., **Jain, P. K.** & **Jain, S. C.** 2009a Important issues in tolerance design of mechanical assemblies. Part 1: tolerance analysis. *Proceedings of the Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture* **223**(10), 1225–1247; doi:10.1243/09544054JEM1304A.

Singh, P. K., **Jain, P. K.** & **Jain, S. C.** 2009b Important issues in tolerance design of mechanical assemblies. Part 2: tolerance synthesis. *Proceedings of the Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture* **223**(10), 1249–1287; doi:10.1243/09544054JEM1304B.

Smith, R. 1997 The historical roots of concurrent engineering fundamentals. *IEEE Transactions on Engineering Management* **44**(1), 67–78; doi:10.1109/17.552809.

Söderberg, R., **Lindkvist, L.** & **Carlson, J.** 2006 Virtual geometry assurance for effective product realization. In *1st Nordic Conference on Product Lifecycle Management - NordPLM'06*, pp. 75–88. Gothenborg, Sweden.

Söderberg, R., **Lindkvist, L.**, **Wärmefjord, K.** & **Carlson, J. S.** 2016 Virtual geometry assurance process and toolbox. *Procedia CIRP* **43**, 3–12; doi:10.1016/j.procir.2016.02.043.

Stewart, S., **Giambalvo, J.**, **Vance, J.**, **Faludi, J.** & **Hoffenson, S.** 2020 A product development approach advisor for navigating common design methods, processes, and environments. *Designs* **4**(1), 4; doi:10.3390/designs4010004.

Stockinger, A. 2010 Computer aided robust design – Verknüpfung rechnerunterstützter Entwicklung und virtueller Fertigung als Baustein des Toleranzmanagements. *Friedrich-Alexander-Universität Erlangen-Nürnberg. PhD thesis. Fortschritt-Berichte VDI Reihe 1, Konstruktionstechnik, Maschinenelemente Nr.* 409.

Stylidis, K., **Wickman, C.** & **Söderberg, R.** 2020 Perceived quality of products: a framework and attributes ranking method. *Journal of Engineering Design* **31**(1), 37–67; doi:10.1080/09544828.2019.1669769.

Sudarsan, R., **Roy, U.**, **Narahari, Y.**, **Sriram, R.**, **Lyons, K. W.** & **Pramanik, N.** 2000 *Information Models for Design Tolerancing: From Conceptual to the Detail Design*, Technical Report NIST IR 6524, National Institute of Standards and Technology, Gaithersburg, MD.

Taguchi, G., **Chowdhury, S.**, **Wu, Y.**, **Taguchi, S.** & **Yano, H.** 2005 *Taguchi's Quality Engineering Handbook*. John Wiley & Sons; ASI Consulting Group.

Teissandier, D., **Delos, V.** & **García, S. C.** 2020 Simulating the respect of a functional condition in a mechanical system with mobilities. *Procedia CIRP* **92**, 106–111; doi:10.1016/j.procir.2020.05.195.

Verband der Automobilindustrie 2013 *Sicherung der Qualität in der Prozesslandschaft: Allgemeines, Reisikoanalysen, Methoden, Vorgehensmodelle*. VDA.

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press

Voelcker, H. B. 1998 The current state of affairs in dimensional tolerancing: 1997. *Integrated Manufacturing Systems* **9**(4), 205–217; doi:10.1108/09576069810217793.

vom Brocke, J., **Hevner, A.** & **Maedche, A.** 2020 Introduction to design science research. In J. vom Brocke, A. Hevner & A. Maedche, eds, *Design Science Research. Cases. Progress in IS*, pp. 1–13. Springer.

Walter, M., **Sprügel, T.** & **Wartzack, S.** 2013 Tolerance analysis of systems in motion taking into account interactions between deviations. *Proceedings of the Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture* **227**(5), 709–719; doi: 10.1177/0954405412473719.

Walter, M., **Storch, M.** & **Wartzack, S.** 2014 On uncertainties in simulations in engineering design: a statistical tolerance analysis application. *Simulation* **90**(5), 547–559; doi: 10.1177/0037549714529834.

Wärmefjord, K., **Hansen, J.** & **Söderberg, R.** 2023 Challenges in geometry assurance of megacasting in the automotive industry. *Journal of Computing and Information Science in Engineering* **23**(6), 060801; doi:10.1115/1.4062269.

Wärmefjord, K., **Söderberg, R.**, **Lindkvist, L.**, **Lindau, B.** & **Carlson, J. S.** 2017 Inspection data to support a digital twin for geometry assurance. In *Volume 2: Advanced Manufacturing*, pp. V002T02A101. American Society of Mechanical Engineers.

Wartzack, S. 2024 *Research in Tolerancing*. Springer.

Wartzack, S., **Meerkamm, H.**, **Stockinger, A.**, **Stoll, T.**, **Stuppy, J.**, **Voß, R.**, **Walter, M.** & **Wittmann, S.** 2011 Lebenszyklusorientierte Toleranzsimulation zur funktionalen und ästhetischen Produktabsicherung. *Konstruktion* **6**, 63–74.

Yu, K. M., **Tan, S. T.** & **Yuen, M. F.** (1994), A review of automatic dimensioning and tolerancing schemes. *Engineering with Computers* **10**(2), 63–80; doi:10.1007/BF01200178.

Zhang, X., **Jiang, K.** & **Ning, R.** (2009), Dimensioning and tolerancing driven by product function model. In *Proceedings of the 11th CIRP Conference on Computer Aided Tolerancing, Annecy, France*.

https://doi.org/10.1017/dsj.2025.6 Published online by Cambridge University Press